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PEE 322 — Past Questions & Materials
by Renance

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Institution: Federal University of Petroleum Resources, Effurun
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Paper 1 of 2 — PEE 322 Well Completion

Level 300 • Second Semester • PastQuestion

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(3) a(i) State and explain the phases of casing design.

(ii). What do you understand by the statement "the detailed design of a casing string"?

(iii) Explain with the aid of a flow diagram, the process involved in the detailed design of a casing string. (8 MARKS)

b. An oil well in Ewereti town in the Niger delta region with 13-3/8" intermediate casing set at 11,750ft, having mud weight of 15ppg. Top of spacer is given to be 2,750ft with spacer weight of 14ppg. top of cement at 3000ft with cement weight 16ppg. Plug bump pressure is 3000psi.

Calculate the net burst load at casing shoe. (7 MARKS)

(4) a Briefly explain the following: (i) Reasons of Cementing an oil well. (ii) Application of Directional Drilling.

(iii) Purposes of installing Casing. (iv) Bumping plug. (v) Types of Casing. (6 MARKS)

4b(i) In a drilling programme done in the North-bank of the Niger Delta region, Calculate the total load of running a 15 3/8", 72 lb/ft intermediate casing to 10,750ft with inside diameter 13.347", mud weight 13ppg, instantaneous velocity 6ft/sec. (7 MARKS)

(ii). What is the purpose of carrying out a casing test. (2 MARKS)

(5) a(i). State and Briefly explain five fundamental properties which must be considered when designing any cement slurry.

(ii). What is Dog leg severity? (ii) List any five cementing equipment and their functions. (8 MARKS).

(iii). The ROP is a function of many factors other than the formation type and formation pressure; state five additional factors.

b. FUPRE 007BIS well was finally drilled as designed, with 9 5/8" production casing set at 11,500 ft with mud weight 11 ppg, top of cement at 7,000 ft with 16 ppg cement weight. Spacer 5,000ft with 13 ppg spacer weight. Calculate the net collapse load at casing shoe and represent it graphically. (7 MARKS)

(6) Determine the cement volume calculations of a 9 5/8" casing of a well to be cemented in place with a single stage cementing operation. The appropriate calculations are to be conducted prior to the operation.

given the details of the operation are as follows: 9 5/8" 47lb/ft casing set at: 12,800'. 12 1/4" hole at 13,810' 13 3/8" 68lb/ft casing set at: 6200'. above shoe. Calculate the total cement slurry. Amount of cement and mix water, Amount of additives and the displacement volume.

Assume gauge hole, add 10% excess in open hole. If the casing is to be cemented with class G cement with the following additives 0.2% D13R (retarder), 1% D65 (friction reducer) slurry density 15.9 ppg. For nat pump 12-p-160, 7" liner 97% eff, 0.138 bbl/sk. (15 MARKS), see Fig 2 below

The casing is to be cemented with class G cement with the following additives:

0.2% D13R (retarder)
1% D65 (friction reducer)
Slurry density = 15.9 ppg

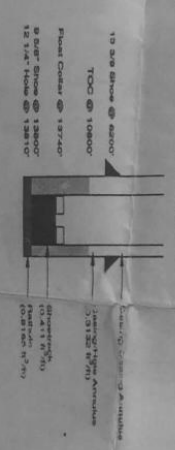
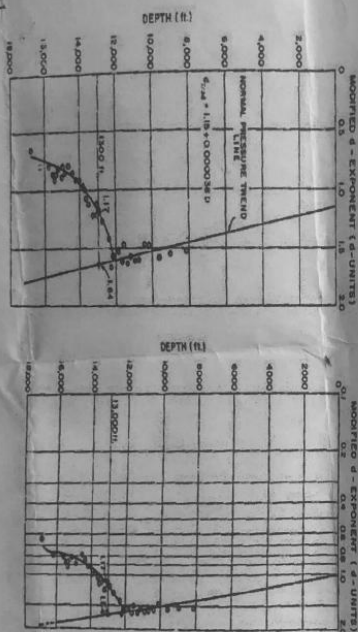


Fig 2.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
SECOND SEMESTER EXAMINATION, 2023/2024 SESSION
PNG 322: WELL COMPLETIONS AND SYSTEM DESIGN
TIME ALLOWED: 3HRS: 11/10/2024. UNIT: 3
ANSWER ANY FIVE QUESTIONS ONLY.

QUESTIONS.

- (1) a. (i) What is well Completion Operation? State and briefly explain its fundamental requirements and classification. (4 MARKS) ✓
(ii) When does the Completion phase begin? (4 MARKS) ✓
b. Briefly explain the function of the following completion components: ✓
Packers, Nipples, Sliding Sleeves, Gas Lift Mandrels and Surface Controlled Subsurface Safety Valves (SCSSV). (4 MARKS)
c. Describe the completion categories listed below, state four advantages and three disadvantages for each. ✓
(i) Cemented and Perforated liner/casing completions (ii) Screen/slotted liner completions (iii) Open hole completions. (7 MARKS)
(2) a. Briefly explain with the aid of a diagram what you understand by: ✓
(i) Formation pressure (ii) Overburden pressure (iii) Formation foreshortening. (6 MARKS) ✓
b. Given depth vs. depth as shown in Figure 1 below. Estimate the formation pressure at 13,000 ft using Retim and McClelland, and the Zamora correlation. (5 MARKS)



c. (i). Identify and explain two ways gas cutting of mud may occur.

(ii). Consider a gas sand which was encountered in the U.S. Gulf Coast area. If the water-filled portion of the sand is pressured normally and the gas/water contact occurred at a depth of 5,000 ft, what mud weight would be required to drill through the top of the sand structure safely at a depth of 4,000 ft? Assume the gas has an average density of 0.76 lbm/gal. (4 MARKS) ✓

Paper 2 of 2 — PEE 322

Level 300 • Second Semester • Handout

Formation Pressures

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Formation Pressures

Learning Outcomes:

Having worked through this chapter the student will be able to:

General:

- Define the terms: Pressure Gradient; hydrostatic pressure; “Normal” Pressure; “Abnormal” Pressure; Overburden(geostatic) Pressure; Fracture pressure.
- Plot the above from a set of data from a well.
- Describe in general terms the origins and mechanisms which generate Overpressured and Underpressured reservoirs
- Describe in detail the mechanism of Undercompaction

- Describe the characteristics of the different types of seal above an abnormally pressured formation and their implications for overpressure detection.
 - Describe the impact of Abnormally pressured formations on well design and drilling operations
- Overpressure Prediction and Detection Techniques:
- List and describe the methods of predicting overpressures before drilling the well. Prioritise these techniques in order of reliability in a given environment.
 - List and describe the techniques used for the detection of overpressures whilst drilling a well.
 - Define the assumptions inherent in, and limitations of, the “d” exponent technique for overpressure detection.
 - Calculate and plot the “d” exponent and modified “d” exponent and use these to determine the top of the overpressured zone.
 - Calculate and plot shale density when used in the determination of overpressures.
- Leak Off Test and Fracture Pressure:
- Describe the mechanisms of formation breakdown
 - Define the terms: Limit test and Leak off test.
 - Describe the procedure used when conducting a leak off test.
 - Calculate the: maximum allowable mudweight (including ECD); and MAASP for the subsequent hole section after conducting a LOT.

1. INTRODUCTION

The magnitude of the pressure in the pores of a formation, known as the formation pore pressure (or

simply formation pressure), is an important consideration in many aspects of well planning and operations. It will influence the casing design and mud weight selection and will increase the chances of

stuck pipe and well control problems. It is particularly important to be able to predict and detect high

pressure zones, where there is the risk of a blow-out.

In addition to predicting the pore pressure in a formation it is also very important to be able to predict the

pressure at which the rocks will fracture. These fractures can result in losses of large volumes of drilling

fluids and, in the case of an influx from a shallow formation, fluids flowing along the fractures all the way

to surface, potentially causing a blowout.

When the pore pressure and fracture pressure for all of the formations to be penetrated have been predicted

the well will be designed, and the operation conducted, such that the pressures in the borehole neither

exceed the fracture pressure, nor fall below the pore pressure in the formations being drilled.

2. FORMATION PORE PRESSURES

During a period of erosion and sedimentation, grains of sediment are continuously building up on top of

each other, generally in a water filled environment. As the thickness of the layer of sediment increases,

the grains of the sediment are packed closer together, and some of the water is expelled from the pore

spaces. However, if the pore throats through the sediment are interconnecting all the way to surface the

pressure of the fluid at any depth in the sediment will be same as that which would be found in a simple

colom of fluid. The pressure in the fluid in the pores of the sediment will only be dependent on the density

of the fluid in the pore space and the depth of the pressure measurement (equal to the height of the colom

of liquid). it will be independent of the pore size or pore throat geometry. The pressure of the fluid in the

pore space (the pore pressure) can be measured and plotted against depth as shown in Figure 1.

This type

of diagram is known as a P-Z diagram The pressure in the formations to be drilled is often expressed in

terms of a pressure gradient. This gradient is derived from a line passing through a particular formation

pore pressure and a datum point at surface and is known as the pore pressure gradient. The reasons for

this will become apparent subsequently. The datum which is generally used during drilling operations is

the drillfloor elevation but a more general datum level, used almost universally, is Mean Sea Level, MSL.

When the pore throats through the sediment are interconnecting, the pressure of the fluid at any depth in

the sediment will be same as that which would be found in a simple colom of fluid and therefore the pore

pressure gradient is a straight line as shown in Figure 1. The gradient of the line is a representation of the

density of the fluid.

Hence the density of the fluid in the pore space is often expressed in units of psi/ft.

This is a very convenient unit of representation since the pore pressure for any given formation can easily

be deduced from the pore pressure gradient if the vertical depth of the formation is known.

Representing

the pore pressures in the formations in terms of pore pressure gradients is also convenient when computing

the density of the drilling fluid that will be required to drill through the formations in question. If the

density of the drilling fluid in the wellbore is also expressed in units of psi/ft then the pressure at all points

in the wellbore can be compared with the pore pressures to ensure that the pressure in the wellbore exceeds

the pore pressure. The differential between the mud pressure and the pore pressure at any given depth is

known as the overbalance pressure at that depth (Figure 2). If the mud pressure is less than the pore

pressure then the differential is known as the underbalance pressure. It will be seen below that the fracture pressure gradient of the formations is also expressed in units of psi/ft.

Most of the fluids found in the pore space of sedimentary formations contain a proportion of salt and are

known as brines. The dissolved salt content may vary from 0 to over 200,000 ppm.

Correspondingly, the

pore pressure gradient ranges from 0.433 psi/ft (pure water) to about 0.50 psi/ft. In most geographical

areas the pore pressure gradient is approximately 0.465 psi/ft (assumes 80,000 ppm salt content) and this

pressure gradient has been defined as the normal pressure gradient.

Any formation pressure above or below the points defined by this gradient are called abnormal pressures

(Figure 3). The mechanisms by which these abnormal pressures can be generated will be discussed below.

When the pore fluids are normally pressured the formation pore pressure is also said to be hydrostatic.

3. OVERBURDEN PRESSURES

The pressures discussed above relate exclusively to the pressure in the pore space of the formations. It is

however also important to be able to quantify the vertical stress at any depth since this pressure will have

a significant impact on the pressure at which the borehole will fracture when exposed to high pressures.

The vertical pressure at any point in the earth is known as the overburden pressure or geostatic pressure.

The overburden gradient is derived from a cross plot of overburden pressure versus depth (Figure 4). The

overburden pressure at any point is a function of the mass of rock and fluid above the point of interest. In

order to calculate the overburden pressure at any point, the average density of the material (rock and fluids)

above the point of interest must be determined. The average density of the rock and fluid in the pore space

is known as the bulk density of the rock :

4. ABNORMAL PRESSURES

Pore pressures which are found to lie above or below the “normal” pore pressure gradient line are called

abnormal pore pressures (Figure 5 and 6). These formation pressures may be either Subnormal (i.e. less

than 0.465 psi/ft) or Overpressured (i.e. greater than 0.465 psi/ft). The mechanisms which generate these

abnormal pore pressures can be quite complex and vary from region to region. However, the most common

mechanism for generating overpressures is called Undercompaction and can be best described by the

undercompaction model.

Casing

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Learning OBJECTIVES

Having worked through this chapter the student will be able to:

General

- State the functions of Casing
- Define the terms: conductor; surface; intermediate; and production casing
- Describe the advantages of using a liner rather than a full string of casing.
- List and describe the loads which must be considered in the design of the casing.

Properties of Casing

- Describe the specific meaning of the terms used to describe the properties of casing: casing size, weight

and grade

- Describe the various types of connection used on casing.

Wellheads and casing hangers

- Describe a conventional wellhead assembly
- Describe the sequence of operations associated with the installation of a spool type wellhead assembly
- Describe a compact spool wellhead and its advantages over the conventional wellhead
- Describe a conventional christmas tree and its function
- Describe the different types of casing hanger that are available and when each would be used.

Casing Running Operations

- Write a step by step program for a casing and liner running and landing operation
- Explain the reasons behind each step in the casing running operation.

Casing Design

- Describe the steps involved in the casing design process.
- Describe the main considerations in selecting the casing size and setting depths.
- Describe and calculate the internal and external loads which are considered when calculating the burst

and collapse loads on a casing.

- Describe the source of tensile loads on casing and the way in which they combine during installation,

cementing and production operations

- Describe the Bi-axial and tri-axial loads which the casing will be subjected to and the way in which these

loads are accommodated in the design process.

1. INTRODUCTION

It is generally not possible to drill a well through all of the formations from surface (or the seabed) to the

target depth in one hole section. The well is therefore drilled in sections, with each section of the well

being sealed off by lining the inside of the borehole with steel pipe, known as casing and filling the annular

space between this casing string and the borehole with cement, before drilling the subsequent hole section.

This casing string is made up of joints of pipe, of approximately 40ft in length, with threaded connections.

Depending on the conditions encountered, 3 or 4 casing strings may be required to reach the target depth.

The cost of the casing can therefore constitute 20-30% of the total cost of the well. Great care must therefore be taken when designing a casing programme which will meet the requirements of the well.

There are many reasons for casing off formations:

- To prevent unstable formations from caving in;
- To protect weak formations from the high mud weights that may be required in subsequent hole sections.

These high mudweights may fracture the weaker zones;

- To isolate zones with abnormally high pore pressure from deeper zones which may be normally pressured;
- To seal off lost circulation zones;

- When set across the production interval: to allow selective access for production / injection/control the

flow of fluids from, or into, the reservoir(s).

One of the casing strings will also be required:

- To provide structural support for the wellhead and BOPs.

Each string of casing must be carefully designed to withstand the anticipated loads to which it will be exposed during installation, when drilling the next hole section, and when producing from the well.

These loads will depend on parameters such as: the types of formation to be drilled; the formation pore

pressures; the formation fracture pressures; the geothermal temperature profile; and the nature of the fluids

in the formations which will be encountered. The designer must also bear in mind the costs of the casing,

the availability of different casing types and the operational problems in running the casing string into the

borehole.

Since the cost of the casing can represent up to 30% of the total cost of the well, the number of casing

strings run into the well should be minimised. Ideally the drilling engineer would drill from surface to

the target depth without setting casing at all. However, it is normally the case that several casing strings

will have to be run into the well in order to reach the objective formations. These strings must be run concentrically with the largest diameter casing being run first and smaller casing strings being used as the well gets deeper. The sizes and setting depths of these casing strings depends almost entirely on the geological and pore pressure conditions in the particular location in which the well is being drilled. Some typical casing string configurations used throughout the world are shown in Figure 1. In view of the high cost of casing, each string must be carefully designed. This design will be based on the anticipated loads to which the casing will be exposed. When drilling a development well, these loads will have been encountered in previous wells and so the casing programme can be designed with a high degree of confidence, and minimal cost. In an exploration well, however, these loads can only be estimated and problems may be encountered which were not expected.

The casing design must therefore be more conservative and include a higher safety margin when quantifying the design loads for which the casing must be designed. In addition, in the case of an exploration well, the casing configuration should be flexible enough to allow an extra string of casing to be run, if necessary. A well drilled in an area with high pressures or troublesome formations will usually require more casing strings than one in a normally pressured environment (Figure 2).

2. Component Parts of a Casing String

A casing string consists of individual joints of steel pipe which are connected together by threaded connections. The joints of casing in a string generally have the same outer diameter and are approximately 40ft long. A bull-nose shaped device, known as a guide shoe or casing shoe, is attached to the bottom of the casing string and a casing hanger, which allows the casing to be suspended from the wellhead, is attached to the top of the casing. Various other items of equipment, associated with the cementing operation, may also be included in the casing string, or attached to the outside of the casing e.g. float collar, centralisers and scratchers. This equipment will be discussed in greater depth in the chapter associated with cementing.

3. Casing Terminology

There are a set of generic terms used to describe casing strings. These terms are shown in Figure 2. The

classification system is based on the specific function of the casing string so, for instance, the function of

the surface string shown in Figure 2 is to support the wellhead and BOP stack. Although there is no direct relationship between the size of casing and its function, there is a great deal of similarity in the

casing sizes used by operators in the North Sea. The chart in Figure 3 shows the most common casing size

and hole size configurations. The dotted lines represent less commonly used configurations. The terms

which are generally used to classify casing strings are shown below. The casing sizes shown alongside

the casing designation are those that are generally used in the Niger Delta and North Sea.

3.1 Conductor Casing (30" O.D.)

The conductor is the first casing string to be run, and consequently has the largest diameter. It is generally

set at approximately 100ft to 500ft below the ground level or seabed.

Its function is to seal off unconsolidated formations at shallow depths which, with continuous mud circulation, would be washed away. The surface formations may also have low fracture strengths which

could easily be exceeded by the hydrostatic pressure exerted by the drilling fluid when drilling a deeper

section of the hole.

In areas where the surface formations are stronger and less likely to be eroded the conductor pipe may not

be necessary. Where conditions are favourable the conductor may be driven into the formation and in this

case the conductor is referred to as a stove pipe.

3.2 Surface Casing (20" O.D.)

The surface casing is run after the conductor and is generally set at approximately 1000 - 1500 ft below

the ground level or the seabed. The main functions of surface casing are to seal off any fresh water sands, and support the wellhead and BOP equipment. The setting depth of this casing string is important in an area where abnormally high pressures are expected. If the casing is set too high, the

formations below the casing may not have sufficient strength to allow the well to be shut-in and killed if

a gas influx occurs when drilling the next hole section. This can result in the formations around the casing

cratering and the influx flowing to surface around the outside of the casing.

3.3 Intermediate Casing (13 3/8" O.D.)

Intermediate (or protection) casing strings are used to isolate troublesome formations between the surface

casing setting depth and the production casing setting depth. The types of problems encountered in this

interval include: unstable shales, lost circulation zones, abnormally pressured zones and squeezing salts.

The number of intermediate casing strings will depend on the number of such problems encountered.

3.4 Production Casing (9 5/8" O.D.)

The production casing is either run through the pay zone, or set just above the pay zone (for an open

hole completion or prior to running a liner). The main purpose of this casing is to isolate the production

interval from other formations (e.g. water bearing sands) and/or act as a conduit for the production tubing. Since it forms the conduit for the well completion, it should be thoroughly pressure tested before

running the completion.

3.5 Liner (7" O.D.)

A liner is a short (usually less than 5000ft) casing string which is suspended from the inside of the previous

casing string by a device known as a liner hanger. The liner hanger is attached to the top joint of the

casing in the string. The liner hanger consists of a collar which has hydraulically or mechanically set slips

(teeth) which, when activated, grip the inside of the previous string of casing. These slips support the

weight of the liner and therefore the liner does not have to extend back up to the wellhead. The overlap

with the previous casing (liner lap) is usually 200ft - 400ft. Liners may be used as an intermediate string or as a production string.

The advantages of running a liner, as opposed to a full string of casing, are that:

- A shorter length of casing string is required, and this results in a significant cost reduction;
- The liner is run on drillpipe, and therefore less rig time is required to run the string;
- The liner can be rotated during cementing operations. This will significantly improve the mud displacement process and the quality of the cement job.

After the liner has been run and cemented it may be necessary to run a casing string of the same diameter

as the liner and connect onto the top of the liner hanger, effectively extending the liner back to surface.

The casing string which is latched onto the top of the liner hanger is called a tie-back string. This tie-

back string may be required to protect the previous casing string from the pressures that will be encountered when the well is in production.

In addition to being used as part of a production string, liners may also be used as an intermediate string

to case off problem zones before reaching the production zone. In this case the liner would be known as a

drilling liner (Figure 2). Liners may also be used as a patch over existing casing for repairing damaged casing or for extra protection against corrosion. In this case the liner is known as a stub liner.

4. PROPERTIES OF CASING

When the casing configuration (casing size and setting depth) has been selected, the loads to which each

string will be exposed will be computed. Casing, of the required size, and with adequate load bearing

capacity will then be selected from manufacturer's catalogues or cementing company handbooks.

Casing joints are manufactured in a wide variety of sizes, weights and material grades and a number of

different types of connection are available. The detailed specification of the sizes, weights and grades of

casing which are most commonly used has been standardised by the American Petroleum Institute - API.

The majority of sizes, weights and grades of casing which are available can be found in manufacturer's

catalogues and cementing company handbooks (e.g. Halliburton Cementing Tables).

Casing is generally classified, in manufacturer's catalogues and handbooks, in terms of its size (O.D.), weight, grade and connection type:

4.1 Casing Size (Outside Diameter - O.D.)

The size of the casing refers to the outside diameter (O.D.) of the main body of the tubular (not the connector). Casing sizes vary from 4.5" to 36" diameter. Tubulars with an O.D. of less than 4.5" are called

Tubing. The sizes of casing used for a particular well will generally be limited to the standard sizes that

are shown in Figure 3. The hole sizes required to accommodate these casing sizes are also shown in this

diagram. The casing string configuration used in any given location e.g. 20" x 13 3/8" x 9 5/8" x 7" x 4

1/2" is generally the result of local convention, and the availability of particular sizes.

4.2 Length of Joint

The length of a joint of casing has been standardised and classified by the API as follows:

Although casing must meet the classification requirements of the API, set out above,

it is not possible to manufacture it to a precise length. Therefore, when the casing is delivered to the rig, the precise length of each joint has to be measured and recorded on a tally sheet. The length is measured from the top of the connector to a reference point on the pin end of the connection at the far end of the casing joint. Lengths are recorded on the tally

sheet to the nearest 100th of a foot. Range 2 is the most common length, although shorter lengths are

useful as pup joints when attempting to assemble a precise length of string.

4.3 Casing Weight

For each casing size there are a range of casing weights available. The weight of the casing is in fact the

weight per foot of the casing and is a representation of the wall thickness of the pipe. There are for instance

four different weights of 9 5/8" casing:

Although there are strict tolerances on the dimensions of casing, set out by the API, the actual I.D. of the

casing will vary slightly in the manufacturing process. For this

reason the drift diameter of casing is quoted in the specifications for all casing.

The drift diameter refers to the guaranteed minimum I.D. of the casing. This may be important when

deciding whether certain drilling or completion tools will be able to pass through the casing e.g. the drift

diameter of 9 5/8" 53.5 lb/ft casing is less than 8 1/2" bit and therefore an 8 1/2" bit cannot be used below

this casing setting depth. If the 47 lb/ft casing is too weak for the particular application then a higher grade

of casing would be used (see below). The nominal I.D. of the casing is used for calculating the volumetric

capacity of the casing.

4.4 Casing Grade

The chemical composition of casing varies widely, and a variety of compositions and treatment processes

are used during the manufacturing process. This means that the physical properties of the steel varies

widely. The materials which result from the manufacturing process have been classified by the API into a

series of "grades" (Table 3). Each grade is designated by a letter, and a number. The letter refers to the

chemical composition of the material and the number refers to the minimum yield strength of the material e.g. N-80 casing has a minimum yield strength of 80000 psi and K-55 has a minimum yield strength of 55000 psi. Hence the grade of the casing provides an indication of the strength of the casing.

The higher the grade, the higher the strength of the casing.

In addition to the API grades, certain manufacturers produce their own grades of material. Both seamless

and welded tubulars are used as casing although seamless casing is the most common type of casing and

only H and J grades are welded.

4.5 Connections

Individual joints of casing are connected together by a threaded connection. These connections are variously classified as: API; premium; gastight; and metal-to metal seal. In the case of API connections, the casing joints are threaded externally at either end and each joint is connected to

the next joint by a coupling which is threaded internally (Figure 5). A coupling is already installed on

one end of each joint when the casing is delivered to the rig. The connection must be leak proof but can

have a higher or lower physical strength than the main body of the casing joint.

A wide variety of threaded connections are available. The standard types of API threaded and coupled connection are:

- Short thread connection (STC)
- Long thread connection (LTC)
- Buttress thread connection (BTC)

In addition to threaded and coupled connections there are also externally and internally upset connections

such as that shown in Figure 4. A standard API upset connection is:

- Extreme line (EL)

The STC thread profile is rounded with 8 threads per inch. The LTC is similar but with a longer coupling, which provides better strength and sealing properties than the STC. The buttress thread profile has flat crests, with the front and back cut at different angles. Extreme line connections also have flat crests and have 5 or 6 threads per inch. The EL connection is the only API connection that has a metal to metal seal at the end of the pin and at the external shoulder of the connection, whereas

all of the other API connections rely upon the thread compound, used to make up the connection, to seal off the leak path between the threads of the connection.

In addition to API connections, various manufacturers have developed and patented their own connections

(e.g. Hydril, Vallourec, Mannesman). These connections are designed to contain high pressure gas and

are often called gastight, premium and metal-to-metal seal connections. These connections are termed

metal-to-metal seal because they have a specific surface machined into both the pin and box of the connection which are brought together and subjected to stress when the connection is made up.

Surveys have shown that over 80% of leaks in casing can be attributed to poor makeup of connections. This may be due to a variety of reasons:

- Excessive torque used in making-up the connections
- Dirty threads
- Cross-threading
- Using the wrong thread compound.

The casing string should be tested for pressure integrity before drilling the subsequent hole section. Most

of the causes of connection failure can be eliminated by good handling and running procedures on the rig.

The recommended make-up torque for API connections is given in API RP 5C1.

These recommended torques are based on an empirical equation obtained from tests using API modified

thread compound on API connections. The recommended make up torque for other connections is available from manufacturers.

5. API Specifications, Standards and Bulletins

The API Committee responsible for the Standardisation of tubular goods is Committee number 5.

This

committee publishes, and continually updates, a series of Specifications, Standards, Bulletins and Recommended Practices covering the manufacture, performance and handling of tubular goods.

The

documents, published by Committee 5, of particular relevance to casing design and specification are :

API SPEC 5CT, "Specification for casing a tubing": Covers seamless and welded casing and tubing, couplings, pup joints and connectors in all grades. Processes of manufacture, chemical and mechanical

property requirements, methods of test and dimensions are included.

API STD 5B, "Specification for threading, gauging, and thread inspection for casing, tubing, and line pipe

threads": Covers dimensional requirements on threads and thread gauges, stipulations on gauging practice,

gauge specifications and certifications, as well as instruments and methods for the inspection of threads

of round-thread casing and tubing, buttress thread casing, and extreme-line casing and drill pipe.

API RP 5A5, "Recommended practice for filed inspection of new casing, tubing and plain-end drill pipe":

Provides a uniform method of inspecting tubular goods.

API RP 5B1, "Recommended practice for thread inspection on casing, tubing and line pipe": The purpose

of this recommended practice is to provide guidance and instructions on the correct use of thread inspection techniques and equipment.

API RP 5C1, "Recommended practice for care and use of casing and tubing": Covers use, transportation,

storage, handling, and reconditioning of casing and tubing.

API RP5C5, "Recommended practice for evaluation procedures for casing and tubing connections":

Describes tests to be performed to determine the galling tendency, sealing performance and structural

integrity of tubular connections.

API BULL 5A2, "Bulletin on thread compounds": Provides material requirements and performance tests

for two grades of thread compound for use on oil-field tubular goods.

API BULL 5C2, "Bulletin on performance properties of casing and tubing": Covers collapsing pressures,

internal yield pressures and joint strengths of casing and tubing and minimum yield load for drill pipe.

API BULL 5C3, "Bulletin on formulas and calculations for casing, tubing, drillpipe and line pipe properties": Provides formulas used in the calculations of various pipe properties, also background information regarding their development and use.

API BULL 5C4, "Bulletin on round thread casing joint strength with combined internal pressure and bending.": Provides joint strength of round thread casing when subject to combined bending and internal pressure.

6. WELLHEADS AND CASING HANGERS

All casing strings, except for liners, are suspended from a wellhead. On a land well or offshore platform the wellhead is just below the rig floor. When drilling offshore, from a floating vessel, the wellhead is installed at the seabed. The wellhead on a land or platform well is made up of a series of

spools, stacked up, one on top of the other (Figure 6).

Surface wellhead spools have four functions:

- To suspend the weight of the casing string;
- To seal off the annulus between successive casing strings at the surface;
- To allow access to the annulus between casing strings;
- To act as an interface between the casing string and BOP stack.

When the casing string has been run into the wellbore it is hung off, or suspended, by a casing hanger,

which rests on a landing shoulder inside the casing spool.

Casing hangers must be designed to take the full weight of the casing, and provide a seal between the

casing hanger and the spool. There are two types of surface wellhead in common use:

6.1 Spool Type Wellhead

The procedure for installing a spool type wellhead system (Figure 6) can be outlined as follows:

(a) The conductor (30") is run and cemented in place. It is then cut off just above the ground level or the

wellhead deck (on a platform);

(b) The 26" hole is drilled and the 20" casing is run through the conductor and cemented.

Sometimes a

landing base is welded onto the top of the 20" casing so that it can rest on the top of the 30" conductor, to

transfer some weight to the 30" casing.

(c) The 20" casing is cut off just above the 30" casing and a 20" casing head housing (lowermost casing

head) is threaded, or welded, onto the top of the casing. The internal profile of this housing has a landing surface on which the casing hanger of the subsequent casing string (13 3/8") lands. The housing has two side outlets which provide access to the 20"x 13 3/8" annulus. The upper flange of the

housing is used as the lower part of the connection to the BOP stack used in drilling the next hole section.

A ring gasket is used to seal off the connection between the housing and the BOP stack.

(d) The 17 1/2" hole is drilled and the 13 3/8" casing is run with the hanger landing in the 20" housing.

The casing is cemented in place. The BOP stack is disconnected and a casing spool (13 5/8") is flanged

up on top of the 20" housing. The BOPs are made up on top of the 13 5/8" spool and the 12 1/4" hole is

drilled.

The process continues, with a new spool being installed for each casing string. Eventually a tubing head

spool is installed. This spool allows the completion tubing to be suspended from the wellhead. The minimum I.D. of a casing spool must be greater than the drift I.D. of the previous casing. A protective

sleeve known as a wear bushing is installed in each spool when it is installed and before the drillstring

is run. The wear bushing must be removed before the next casing string is run.

Finally the Christmas tree is installed on top of the wellhead (Figure 7). A ring gasket, approved by the

API, is used to seal off the space between the flanges on the spools. The gaskets have pressure energized

seals and can be rated up to 15000 psi.

The disadvantages of this type of wellhead are:

- a lot of time is spent flanging up the spools;
- the large number of seals, increases the chance of a pressure leak;
- BOPs must be removed to install the next casing spool;
- a lot of headroom is required, which may not be available in the wellhead area of an offshore platform

6.2 Compact Spool (Speedhead)

The compact spool was developed as an alternative to the conventional spool discussed above. A compact

spool enables several casing strings or tubing to be suspended from a single spool. The first step in using this type of wellhead is to install the 20" casing head housing, as in the case of the spool type wellhead. After the 13 3/8" casing is run and cemented, the casing is cut off and the speedhead is connected to the casing head housing. The BOPs can then be connected to the top of the housing, and the

next hole section drilled.

The 9 5/8" casing is then run, with the hanger resting on a landing shoulder inside the speedhead. A 7"

casing string can be run, and landed, in the speedhead in a similar manner to the 9 5/8" casing. The tubing

string may also be run and landed in the speedhead. The Christmas tree can then be installed on top of the

speedhead. The disadvantage of the compact spool is that the casing programme cannot be easily

altered, and so this system is less flexible than the separate spool system.

6.3 Casing Hangers

There are two types of casing hanger in common use. Wellheads can be designed to accept both types of hanger.

Mandrel (boll weevil) Type Casing Hangers: This type of hanger (Figure 8) is screwed onto the top of the casing string so that it lands in the casing housing when the casing shoe reaches the required depth.

Short lengths of casing, known as pup joints may have to be added to the string so that the casing shoe is at the correct depth when the hanger lands in the wellhead. The calculation which determines the length of pup joints required to achieve this positioning is known as spacing out the string. Although this is the most common type of hanger it cannot be used if there is a risk that the casing will not reach bottom and therefore that the hanger will not land in the wellhead.

Slip Type Casing Hangers: This type of hanger (Figure 9) is wrapped around the casing and then lowered until it sits inside the casing spool. The slips are automatically set when the casing is lowered (in a similar fashion to drillpipe slips). This type of hanger can be used if the casing stands up on a ledge and cannot reach its required setting depth. These types of hanger are also used when tension has to be applied in order to avoid casing buckling when the well is brought into production

7. RIG-SITE OPERATIONS

Casing leaks are often caused by damaging the threads while handling and running the casing on the rig.

It has also been known for a joint of the wrong weight or grade of casing to be run in the wrong place, thus creating a weak spot in the string.

Such mistakes are usually very expensive to repair, both in terms of rig time and materials. It is important, therefore, to use the correct procedures when running the casing.

7.1 Handling Procedures

(a) When the casing arrives at the rig site the casing should be carefully stacked in the correct running order. This is especially important when the string contains sections of different casing grades and weights.

On offshore rigs, where deck space is limited, do not stack the casing too high or else excessive lateral

loads will be imposed on the lowermost row. Casing is off-loaded from the supply boat in reverse order,

so that it is stacked in the correct running order.

(b) The length, grade, weight and connection of each joint should be checked and each joint should be

clearly numbered with paint. The length of each joint of casing is recorded on a tally sheet. If any joint is

found to have damaged threads it can be crossed off the tally sheet. The tally sheet is used by the Drilling

engineer to select those joints that must be run so that the casing shoe ends up at the correct depth when

the casing hanger is landed in the wellhead.

(c) While the casing is on the racks the threads and couplings should be thoroughly checked and cleaned.

Any loose couplings should be tightened (d) Casing should always be handled with thread protectors in

place. These need not be removed until the joint is ready to be stabbed into the string.

7.2 Casing Running Procedures

(a) Before the casing is run, a check trip should be made to ensure that there are no tight spots or ledges

which may obstruct the casing and prevent it reaching bottom.

(b) The drift I.D. of each joint should be checked before it is run.

(c) Joints are picked up from the catwalk and temporarily rested on the ramp. A single joint elevator is

used to lift the joint up through the "V" door into the derrick (Figure 10).

(d) A service company (casing crew) is usually hired to provide a stabber and one or two floormen to

operate the power tongs. The stabbing board is positioned at the correct height to allow the stabber to

centralise the joint directly above the box of the joint suspended in the rotary table. The pin is then carefully stabbed into the box and the power tongs are used to make up the connection slowly to ensure

that the threads of the casing are not cross threaded. Care should be taken to use the correct thread

compound to give a good seal. The correct torque is also important and can be monitored from a torque

gauge on the power tongs. On buttress casing there is a triangle stamped on the pin end as a reference

mark. The coupling should be made up to the base of the triangle to indicate the correct make-up.

(e) As more joints are added to the string the increased weight may require the use of heavy duty slips

(spider) and elevators (Figure 11).

(f) If the casing is run too quickly into the hole, surge pressures may be generated below the casing in the

open hole, increasing the risk of formation fracture. A running speed of 1000 ft per hour is often used in

open hole sections. If the casing is run with a float shoe the casing should be filled up regularly as it is

run, or the casing will become buoyant and may even collapse, under the pressure from the mud in the

hole.

(g) The casing shoe is usually set 10-30 ft off bottom.

7.3 Casing Landing Procedures

After the casing is run to the required depth it is cemented in place while suspended in the wellhead. The

method used for landing the casing will vary from area to area, depending on the forces exerted on the

casing string after the well is completed.

These forces may be due to changes in formation pressure, temperature, fluid density and earth movements

(compaction). These will cause the casing to either shrink or expand, and the landing procedure must take

account of this.

There are basically 3 different ways in which the casing can be cemented and landed:

- landing the casing and cementing;
- suspending the casing, conducting the cement job and then applying additional tension when the cement has hardened;
- landing the casing under compression;

The first case does not require any action after the cementing operation is complete.

The casing is simply landed on a boll-weevil hanger and cemented in place. Additional tension (over and

above the suspended weight) may however have to be applied to the casing to prevent buckling due to

thermal expansion when the well is producing hot fluids. Additional tension can be applied, after the

casing has been cemented, by suspending the casing from the elevators during the cementing operation

and then applying an overpull (extra tension) to the casing once the cement has hardened.

The casing would then be landed on a slip and seal assembly. The level of overpull applied to the casing

will depend on the amount of buckling load that is anticipated due to production. The third option may be

required in the case that the suspended tension reduces the casing's collapse resistance below an acceptable level. In this case the casing is suspended from the elevators during cementing and then lowered

until the desired compression is achieved before setting the slip and seal assembly.

7.4 Liner Running Procedures

Liners are run on drillpipe with special tools which allow the liner to be run, set and cemented all in one

trip (Figure 12). The liner hanger is installed at the top of the liner. The hanger has wedge slips which can

be set against the inside of the previous string. The slips can be set mechanically (rotating the drillpipe)

or hydraulically (differential pressure). A liner packer may be used at the top of the liner to seal off the

annulus after the liner has been cemented.

The basic liner running procedure is as follows:

- (a) Run the liner on drillpipe to the required depth;
- (b) Set the liner hanger;
- (c) Circulate drilling fluid to clean out the liner;
- (d) Back off (disconnect) the liner hanger setting tool;
- (e) Pump down and displace the cement;
- (f) Set the liner packer;
- (g) Pick up the setting tool, reverse circulate to clean out cement and pull out of hole.

8. CASING DESIGN

8.1 Introduction to the Casing Design Process

The casing design process involves three distinct operations: the selection of the casing sizes and setting

depths; the definition of the operational scenarios which will result in burst, collapse and axial loads being

applied to the casing; and finally the calculation of the magnitude of these loads and selection of an appropriate weight and grade of casing. The steps in the casing design process are shown in Figure 13.

8.1.1 Design Casing Scheme Configuration - Select Casing sizes and Setting Depths

The casing setting depths are selected on the basis of an assessment of the conditions to be encountered

when drilling the subsequent hole section or, in the case of production casing, the completion design.

The first step in deciding upon the setting depth for the surface and intermediate casing strings is to

calculate the maximum pressures that could be encountered in the hole section below the string in question. These pressures must not exceed the formation strength at any point in the hole and in particular

at the casing shoe. The highest pressure that will be encountered in the open hole section will occur when

circulating out a gas influx (see chapter on Well Control). The formation strength can be estimated from

nearby well data or by calculation (see chapter on Formation Pressures and Fracture Strength).

The procedure for establishing the acceptable setting depth is illustrated in Figure

1. Start at Total Depth (TD) of the Well

2. Determine the formation fracture pressure at all points in the well
3. Calculate the borehole pressure profile when circulating out a gas influx from TD
4. Plot the formation fracture pressure and the wellbore pressure when circulating out an influx, on the same axes
5. The casing must be set at least at the depth where the two plots cross i.e. this is the shallowest depth at which the casing can be safely set. If the casing is set any shallower when drilling this hole section then the formation will fracture if an influx occurs.
6. Repeat steps 2 to 5 moving up the well, with each subsequent string starting at the casing setting depth for each string.

The setting depth of the casing will also be determined by a range of other considerations such as: the

need to isolate weak formations from high mud weights; isolate lost circulation zones; and to isolate

troublesome formations, such as shales, which can cause hole problems whilst drilling subsequent formations.

The casing sizes and string configuration are dictated by the size of the smallest casing string to be run in

hole. Once the smallest casing size is known all subsequent casing sizes (and hole sizes) are selected from

Figure 3. The smallest casing size is selected on the basis of operational considerations such as: the size

and configuration of the completion string or well testing and/or the size of the logging tools to be run

through the casing. The drilling engineer will collate this information from the geology, reservoir engineering and production engineering departments. The objective of the drilling engineer is to use the

smallest casing sizes possible. It can be readily appreciated that if it is acceptable to use a 4" casing string

as the production casing then the next string will be 7", the next 9 5/8" and so forth. Hence, if only three

casing strings are required then the surface string can be 9 5/8". This slimhole design will result in considerable savings in drilling and equipment costs.

8.1.2 Define the Operational Scenarios and Consequent Loads on the Casing

The loads to which the casing will be exposed during the life of the well will depend on the operations to

be conducted: whilst running the casing; drilling the subsequent hole section; and during the producing

life of the well. These operations will result in radial (burst and collapse) and axial (tensile and compressive) loads on the casing strings. Since the operations conducted inside any particular string (e.g.

the surface string) will differ from those inside the other strings (e.g. the production string) the load scenarios and consequent loads will be specific to a particular string.

The definition of the operational scenarios to be considered is one of the most important steps in the casing

design process and they will therefore generally be established as a company policy.

8.1.3 Calculate the Loads on the Casing and Select the Appropriate Weight and Grade of Casing

Having defined the size and setting depth for the casing strings, and defined the operational scenarios to

be considered, the loads to which the casing will be exposed can be computed. The particular weight and

grade of casing required to withstand these loads can then be determined.

The uniaxial loads to which the casing is exposed are:

Collapse Load

The casing will experience a net collapse loading if the external radial load exceeds the internal radial load

(Figure 15). The greatest collapse load on the casing will occur if the casing is evacuated (empty) for any

reason. The collapse load, P_c at any point along the casing can be calculated from:

Burst Load

The casing will experience a net burst loading if the internal radial load exceeds the external radial load.

The burst load, P_b at any point along the casing can be calculated from:

$$P_b = P_i - P_e$$

In designing the casing to resist burst loading the pressure rating of the wellhead and BOP stack should

be considered since the casing is part of the well control system.

The internal, P_i and external, P_e loads which are used in the determination of the burst and collapse loads

on the casing are derived from an analysis of operational scenarios.

External Loads, P_e :

The following issues are considered when deciding upon the external load to which the casing will be

subjected:

(a.) The pore pressure in the formation (pore pressure)

If the engineer is satisfied that it will be possible to displace all of the mud from the annulus between the

casing and borehole during the cementing operation, and that a satisfactory cement sheath can be achieved,

the formation pore pressure is generally used to determine the load acting on the casing below the top of

cement in the annulus, after the cement has hardened.

(b.) The weight of the mud in which the casing was run.

If a poor cement bond between the casing and cement or cement and borehole is anticipated then the

pressure due to a column of mud in the annulus is generally used to determine the load acting on the casing

below the top of cement in the annulus, after the cement has hardened. If the mud has been in place for

more than 1 year the weighting material will probably have settled out and therefore the pressure experienced by the casing will be due to a column of mud mixwater (water or base-oil).

(c.) The pressure from a column of cement mixwater

The pressure due to the cement mixwater is often used to determine the external load on the casing during

the producing life of the well. This pressure is equal to the density of fresh or seawater in the case of

water-based mud and base oil in the case of oil based mud. The assumption is that the weighting material

in the mud (generally Barite) has settled from suspension.

(d.) The pressure due to a column of cement slurry

The pressure exerted by a column of cement slurry will be experienced by the casing until the cement sets.

It is assumed that hardened cement does not exert a hydrostatic pressure on the casing.

(e.) Blockage in the annulus

If a blockage of the annulus occurs during a stinger cement operations (generally performed on a conductor

casing). The excess pumping pressure on the cement will be transmitted to the annulus but not to the inside

of the casing. This will result in an additional external load during stinger cementing. In the case of conventional cementing operations a blockage in the annulus will result in an equal and opposite pressures

inside and outside the casing.

Internal Loads, P_i :

It is common place to consider the internal loads due to the following:

(a.) Mud to Surface:

This will be the predominant internal pressure during drilling operations. The casing designer must consider the possibility that the density of the drilling fluid may change during the drilling operation, due

to for instance lost circulation or an influx.

(b.) Pressure due to influx

The worst case scenario which can arise, from the point of view of burst loading, is if an influx of hydrocarbons occurs, that the well is completely evacuated to gas and simultaneously closed in at the BOP

stack.

(c.) Full Evacuation

The worst case scenario which can arise, from the point of view of collapse loading,

is if the casing is completely evacuated.

(d.) Production Tubing Leak

In the case of production casing specifically a leak in the production tubing will result in the tubing pressure being exposed to the casing. The closed in tubing pressure is used as the basis of determining the

pressure on the casing. This is calculated on the basis of a column of gas against the formation pressure.

The pressure below surface is based on the combined effect of the tubing head pressure and the hydrostatic

pressure due to a column of packer fluid (if there is any in the annulus).

(e.) Fracture Pressure of Open Formations

When considering the internal loads on a casing string the fracture pressure in any formations open to the

internal pressures must be considered. The pressure in the open hole section cannot exceed the fracture

pressure of the weakest formation.

Hence, the pressures in the remaining portion of the borehole and the casing will be controlled by this

fracture pressure. The formation just below the casing shoe is generally considered to be the weakest

formation in the open hole section.

Net Radial Loading (Burst or Collapse Load)

When the internal and external loads have been quantified the maximum net radial loading on the casing

is determined by quantifying the difference between the internal and external load at all points along the

casing. If the net radial loading is outward then the casing is subjected to a burst load. If the net loading is

inward then the casing is subjected to a collapse load. The internal and external loads used in the determination of the net load must be operationally compatible i.e. it must be possible for them to co-exist

simultaneously.

Axial Load

The axial load on the casing can be either tensile or compressive, depending on the operating conditions

(Figure 16). The axial load on the casing will vary along the length of the casing. The casing is subjected

to a wide range of axial loads during installation and subsequent drilling and production. The axial loads

which will arise during any particular operation must be computed and added together to determine the

total axial load on the casing.

Design Factors

The uncertainty associated with the conditions used in the calculation of the external, internal, compressive

and tensile loads described above is accommodated by increasing the burst collapse and axial loads by a

Design Factor. These factors are applied to increase the actual loading figures to obtain the design loadings.

Design factors are determined largely through experience, and are influenced by the consequences of a

casing failure. The degree of uncertainty must also be considered (e.g. an exploration well may require

higher design factors than a development well),

The following ranges of factors are commonly used:

- Burst design factors 1.0 - 1.33
- Collapse design factors 1.0 - 1.125
- Tension design factors 1.0 - 2.0
- Triaxial Design Factors 1.25

8.2 Casing Design Rules Base

The loading scenarios to be used in the design of the casing string will be dictated by the operating company, on the basis of international and regional experience. These loading scenarios are generally

classified on the basis of the casing string classification. The following rules base is presented as a typical

example of a casing design rules base.

When the load case has been selected the internal and external loads are calculated on the basis of the

rules below. These loads are then plotted on a common axis and the net loading (burst or collapse) is

computed. An appropriate casing string can then be selected from the casing tables.

Conductor:

The predominant concern in terms of failure of the conductor casing during installation is collapse of the

casing. Whilst running the casing it is highly unlikely that the casing will be subjected to a differential

pressure. When conducting the cement job the inside of the casing will generally contain the drilling fluid

in which the casing was run into the well. The maximum external load will be due to the borehole-casing

annulus being full of cement (assumes cement to surface). If a stab in tinger cementation job is conducted

there is the possibility that the annulus will bridge off during the cementing operation and since this pressure will be isolated from the annulus between the casing and the drillpipe stinger this pressure will

not be experienced on the inside of the casing. Hence, very high collapse loads will be experienced by the

casing below the point at which the bridging occurs.

The design scenario to be used for collapse of conductors in this course (and the examinations) is when the casing is fully evacuated due to lost circulation whilst drilling. In this case the casing is empty on the inside and the pore pressure is acting on the outside.

The maximum burst load is experienced if the well is closed in after a gas kick has been experienced.

The pressure inside the casing is due to formation pore pressure at the bottom of the well and a column of gas which extends from the bottom of the well to surface. It is assumed that pore pressure is acting on the outside of the casing.

Note that it would be very unusual to close a well in due to a "shallow" kick below the conductor. It would be more common to allow the influx to flow to surface and divert it away from the rig. This is to avoid the possibility of the formation below the shoe fracturing.

Surface Casing:

Once the surface casing has been set a BOP stack will be placed on the wellhead and in the event of a kick

the well will be closed in at surface and the kick circulated out of the well. The surface casing must therefore be able to withstand the burst loads which will result from this operation. Some operators will

require that the casing be designed to withstand the burst pressures which would result from internal

pressures due to full evacuation of the well to gas.

The maximum collapse loads may be experienced during the cement operation or due to lost circulation

whilst drilling ahead.

The design scenario to be used for collapse of surface casing in this course (and the examinations) is when the casing is fully evacuated due to lost circulation whilst drilling. In this case the casing is empty on the inside and the pore pressure is acting on the outside.

The maximum burst load is experienced if the well is closed in after a gas kick has been experienced.

The pressure inside the casing is due to formation pore pressure at the bottom of the well and a column of gas which extends from the bottom of the well to surface. It is assumed that pore pressure is acting on the outside of the casing

Intermediate Casing:

The intermediate casing is subjected to similar loads to the surface casing.

The design scenario to be used for collapse of intermediate casing in this course (and the examinations) is when the casing is fully evacuated due to lost circulation whilst drilling. In this case

the casing is empty on the inside and the pore pressure is acting on the outside.

The maximum burst load is experienced if the well is closed in after a gas kick has been experienced.

The pressure inside the casing is due to formation pore pressure at the bottom of the well and a column of gas which extends from the bottom of the well to surface. It is assumed that pore pressure

is acting on the outside of the casing.

Production Casing:

The design scenarios for burst and collapse of the production casing are based on production operations.

The design scenario to be used for burst of production casing in this course (and the examinations) is when a leak is experienced in the tubing just below the tubing hanger. In this event the pressure at the top of the casing will be the result of the reservoir pressure minus the pressure due to a column

of gas. This pressure will then act on the fluid in the annulus of well and exert a very high internal pressure at the bottom of the casing.

The design scenario to be used for collapse of production casing in this course (and the examinations) is when the annulus between the tubing and casing has been evacuated due to say the use of gaslift.

8.3 Other design considerations

In the previous sections the general approach to casing design has been explained. However, there are

special circumstances which cannot be satisfied by this general procedure. When dealing with these cases

a careful evaluation must be made and the design procedure modified accordingly. These special circumstances include:

- Temperature effects - high temperatures will tend to expand the pipe, causing buckling. This must be

considered in geothermal wells.

- Casing through salt zones - massive salt formations can flow under temperature and pressure. This will

exert extra collapse pressure on the casing and cause it to shear. A collapse load of around 1 psi/ft (overburden stress) should be used for design purposes where such a formation is present.

- Casing through H₂S zones - if hydrogen sulphide is present in the formation it may cause casing failures

due to hydrogen embrittlement. L-80 grade casing is specially manufactured for use in H₂S zones

8.4 Summary of Design Process

The design process can be summarised as follows:

1. Select the Casing sizes and setting depths on the basis of: the geological and pore pressure prognosis

provided by the geologist and reservoir engineer; and the production tubing requirements on the basis of

the anticipated productivity of the formations to be penetrated.

2. Define the operational scenarios to be considered during the design of each of the casing strings. This

should include installation, drilling and production (as appropriate) operations.

3. Calculate the burst loading on the particular casing under consideration.

4. Calculate the collapse loading on the particular casing under consideration.

5. Increase the calculated burst and collapse loads by the Design Factor which is appropriate to the casing

type and load conditions considered.

6. Select the weight and grade of casing (from manufacturers tables or service company tables) which

meets the load conditions calculated above.

7. For the casing chosen, calculate the axial loading on the casing. Apply the design factor for the casing

and load conditions considered and check that the pipe body yield strength of the selected casing exceeds

the axial design loading. Choose a coupling whose joint strength is greater than the design loading. Select

the same type of coupling throughout the entire string.

8. Taking the actual tensile loading from 7 above determine the reduction in collapse resistance at the top

and bottom of the casing.

Several attempts may have to be made before all these loading criteria are satisfied and a final design is

produced. When deciding on a final design bear the following points in mind:

- Include only those types of casing which you know are available. In practice only a few weights and

grades will be kept in stock.

- Check that the final design meets all requirements and state clearly all design assumptions.

- If several different designs are possible, choose the most economical scheme that meets requirements.

Cementing 8

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Cementing

Learning OBJECTIVES :

Having worked through this chapter the student will be able to:

General

- Describe the principal functions of cement.

Cement Slurries

- List and describe the major properties of a cement slurry.
- Describe the additives used in cement slurries and the way in which they affect the properties of the slurry.

Cementing Operations

- Calculate the volume of : slurry, cement, mixwater, displacing fluid required for a single stage and two-stage cementing operation.
- Calculate the bottomhole pressures generated during the above cementing operations.
- Describe the surface and downhole equipment used in a single, two-stage and liner cementation operation.
- Prepare a program for a single and two stage cementing operation and describe the ways in which a good cement bond can be achieved.

Cement Plugs

- Describe the reasons for setting cement plugs.
- Describe the principal methods for placing a cement plug in casing or open hole.
- Calculate the displacement volumes for an underbalanced cement plug.

Evaluation of Cementing Operations

- Describe the principles involved and the tools and techniques used to evaluate the quality of a cementing operation.
- Discuss the limitations of the above techniques.

Cementing

1. Introduction

Cement is used primarily as an impermeable seal material in oil and gas well drilling. It is most widely

used as a seal between casing and the borehole, bonding the casing to the formation and providing a barrier

to the flow of fluids from, or into, the formations behind the casing and from, and into, the subsequent

hole section (Figure

1). Cement is also used for remedial or repair work on producing wells.

It is used for instance to seal off perforated casing when a producing zone starts to produce large amounts

of water and/or to repair casing leaks.

(This chapter will present: the reasons for using cement in oil and gas well drilling; the design of the cement slurry; and the operations involved in the placement of the cement slurry. The methods used to determine if the cementing operation has been successful will also be discussed.)

1.1 Functions of oilwell cement

There are many reasons for using cement in oil and gas well operations. As stated above, cement is most

widely used as a seal between casing and the borehole, bonding the casing to the formation and providing a barrier to the flow of fluids from, or into, the formations behind the casing and from, and into,

the subsequent hole section (Figure 1). However, when placed between the casing and borehole the cement

may be required to perform some other tasks.

The most important functions of a cement sheath between the casing and borehole are:

- To prevent the movement of fluids from one formation to another or from the formations to surface

through the annulus between the casing and borehole.

- To support the casing string (specifically surface casing)
- To protect the casing from corrosive fluids in the formations.

However, the prevention of fluid migration is by far the most important function of the cement sheath

between the casing and borehole. Cement is only required to support the casing in the case of the surface

casing where the axial loads on the casing, due to the weight of the wellhead and BOP connected to the

top of the casing string, are extremely high. The cement sheath in this case prevents the casing from

buckling.

The techniques used to place the cement in the annular space is done by pumping cement down the inside

of the casing and through the casing shoe into the annulus (Figure 2). This operation is known as a primary cement job. A successful primary cement job is essential to allow further drilling and production

operations to proceed.

Another type of cement job that is performed in oil and gas well operations is called a secondary or squeeze cement job. This type of cement job may have to be done at a later stage in the life of the well.

A secondary cement job may be performed for many reasons, but is usually carried out on wells which

have been producing for some time. They are generally part of remedial work on the well (e.g. sealing off

water producing zones or repairing casing leaks). These cement jobs are often called squeeze cement jobs

because they involve cement being forced through holes or perforations in the casing into the annulus

and/or the formation (Figure 3).

The specific properties of the cement slurry which is used in the primary and secondary cementing operations discussed above will depend on the particular reason for using the cement (e.g. to plug off the

entire wellbore or simply to plug off perforations) and the conditions under which it will be used (e.g. the

pressure and temperature at the bottom of the well).

The cement slurry which is used in the above operations is made up from: cement powder; water; and

chemical additives. There are many different grades of cement powder manufactured and each has particular attributes which make it suitable for a particular type of operation. These grades of cement

powder will be discussed below. The water used may be fresh or salt water. The chemical additives (Figure

4) which are mixed into the cement slurry alter the properties of both the cement slurry and the hardened

cement and will be discussed at length in Section 3 below.

Each cement job must be carefully planned to ensure that the correct cement and additives are being used,

and that a suitable placement technique is being employed for that particular application. In planning the

cement job the engineer must ensure that:

- The cement can be placed correctly using the equipment available
- The cement will achieve adequate compressive strength soon after it is placed
- The cement will thereafter isolate zones and support the casing throughout the life of the well

To assist the engineer in designing the cement slurry, the cement slurry is tested in the laboratory under

the conditions to which it will be exposed in the wellbore.

These tests are known as pilot tests and are carried out before the job goes ahead. These tests must simulate

downhole conditions as closely as possible. They will help to assess the effect of different amounts of

additives on the properties of the cement (e.g. thickening time, compressive strength development etc).

1.2 Classification of cement powders

There are several classes of cement powder which are approved for oilwell drilling applications, by the

American Petroleum Institute - API. Each of these cement powders have different properties when mixed

with water. The difference in properties produced by the cement powders is caused by the differences in

the distribution of the four basic compounds which are used to make cement powder; C3S, C2S, C3A,

C4AF (Table 1).

Classes A and B - These cements are generally cheaper than other classes of cement and can only be used

at shallow depths ,where there are no special requirements.

Class B has a higher resistance to sulphate than Class A.

Class C - This cement has a high C3S content and therefore becomes hard relatively quickly.

Classes D,E and F - These are known as retarded cements since they take a much longer time to set hard

than the other classes of cement powder. This retardation is due to a coarser grind. These cement powders

are however more expensive than the other classes of cement and their increased cost must be justified by

their ability to work satisfactorily in deep wells at higher temperatures and pressures.

Class G and H - These are general purpose cement powders which are compatible with most additives

and can be used over a wide range of temperature and pressure.

Class G is the most common type of cement and is used in most areas . Class H has a coarser grind than

Class G and gives better retarding properties in deeper wells.

There are other, non-API, terms used to classify cement. These include the following:

- Pozmix cement - This is formed by mixing Portland cement with pozzolan (ground volcanic ash) and

2% bentonite. This is a very lightweight but durable cement. Pozmix cement is less expensive than most

other types of cement and due to its light weight is often used for shallow well casing cementation operations.

- Gypsum Cement - This type of cement is formed by mixing Portland cement with gypsum. These cements develop a high early strength and can be used for remedial work. They expand on setting and

deteriorate in the presence of water and are therefore useful for sealing off lost circulation zones.

- Diesel oil cement - This is a mixture of one of the basic cement classes (A, B, G, H), diesel oil or kerosene and a surfactant. These cements have unlimited setting times and will only set in the presence of

water. Consequently they are often used to seal off water producing zones, where they absorb and set to

form a dense hard cement.

1.3 Mixwater Requirements

The water which is used to make up the cement slurry is known as the mixwater. The amount of mixwater

used to make up the cement slurry is shown in Table 2.

These amounts are based on :

- The need to have a slurry that is easily pumped.
- The need to hydrate all of the cement powder so that a high quality hardened cement is produced.
- The need to ensure that all of the free water is used to hydrate the cement powder and that no free water

is present in the hardened cement.

The amount of mixwater that is used to make up the cement slurry is carefully controlled. If too much

mixwater is used the cement will not set into a strong, impermeable cement barrier. If not enough mixwater is used :

- The slurry density and viscosity will increase.
- The pumpability will decrease
- Less volume of slurry will be obtained from each sack of cement

The quantities of mixwater quoted in Table 2 are average values for the different classes of cement.

Sometimes the amount of mixwater used will be changed to meet the specific temperature and pressure

conditions which will be experienced during the cement job.

2. PROPERTIES OF CEMENT

The properties of a specific cement slurry will depend on the particular reason for using the cement, as

discussed above. However, there are fundamental properties which must be considered when designing

any cement slurry.

(a) Compressive strength

The casing shoe should not be drilled out until the cement sheath has reached a compressive strength of

about 500 psi. This is generally considered to be enough to support a casing string and to allow drilling to

proceed without the hardened cement sheath, disintegrating, due to vibration. If the operation is delayed

whilst waiting on the cement to set and develop this compressive strength the drilling rig is said to be

“waiting on cement” (WOC). The development of compressive strength is a function of several variables, such as: temperature; pressure; amount of mixwater added; and elapsed time since mixing.

The setting time of a cement slurry can be controlled with chemical additives, known as accelerators.

Table 3 shows the compressive strengths for different cements under varying conditions.

(b) Thickening time (pumpability)

The thickening time of a cement slurry is the time during which the cement slurry can be pumped and

displaced into the annulus (i.e. the slurry is pumpable during this time). The slurry should have sufficient

thickening time to allow it to be:

- Mixed
- Pumped into the casing
- Displaced by drilling fluid until it is in the required place

Generally 2 - 3 hours thickening time is enough to allow the above operations to be completed. This also

allows enough time for any delays and interruptions in the cementing operation. The thickening time that

is required for a particular operation will be carefully selected so that the following operational issues are

satisfied:

- The cement slurry does not set whilst it is being pumped
- The cement slurry is not sitting in position as a slurry for long periods, potentially being contaminated

by the formation fluids or other contaminants

- The rig is not waiting on cement for long periods.

Wellbore conditions have a significant effect on thickening time. An increase in temperature, pressure or

fluid loss will each reduce the thickening time and these conditions will be simulated when the cement

slurry is being formulated and tested in the laboratory before the operation is performed.

(c) Slurry density

The standard slurry densities shown in Table 2 may have to be altered to meet specific operational requirements (e.g. a low strength formation may not be able to support the hydrostatic pressure of a cement

slurry whose density is around 15 ppg). The density can be altered by changing the amount of mixwater

or using additives to the cement slurry. Most slurry densities vary between 11 - 18.5 ppg.

It should be noted that these densities are relatively high when the normal formation pore pressure gradient

is generally considered to be equivalent to 8.9 ppg. It is generally the case that cement slurries generally

have a much higher density than the drilling fluids which are being used to drill the well. The high slurry

densities are however unavoidable if a hardened cement with a high compressive strength is to be achieved.

(d) Water loss

The slurry setting process is the result of the cement powder being hydrated by the mixwater. If water is

lost from the cement slurry before it reaches its intended position in the annulus its pumpability will

decrease and water sensitive formations may be adversely affected. The amount of water loss that can be

tolerated depends on the type of cement job and the cement slurry formulation.

Squeeze cementing requires a low water loss since the cement must be squeezed before the filter cake

builds up and blocks the perforations. Primary cementing is not so critically dependent on fluid loss. The

amount of fluid loss from a particular slurry should be determined from laboratory tests. Under standard

laboratory conditions (1000 psi filter pressure, with a 325 mesh filter) a slurry for a squeeze job should

give a fluid loss of 50 - 200 cc. For a primary cement job 250 - 400 cc is adequate.

(e) Corrosion resistance

Formation water contains certain corrosive elements which may cause deterioration of the cement sheath.

Two compounds which are commonly found in formation waters are sodium sulphate and magnesium

sulphate. These will react with lime and C3S to form large crystals of calcium sulphotoaluminate. These

crystals expand and cause cracks to develop in the cement structure. Lowering the C3A content of the

cement increases the sulphate resistance. For high sulphate resistant cement the C3A content should be 0

- 3%

(f) Permeability

After the cement has hardened the permeability is very low (<0.1 millidarcy). This is much lower than

most producing formations. However if the cement is disturbed during setting (e.g. by gas intrusion) higher

permeability channels (5 - 10 darcies) may be created during the placement operation.

3. CEMENT ADDITIVES

Most cement slurries will contain some additives, to modify the properties of the slurry and optimise the

cement job. Most additives are known by the trade-names used by the cement service companies. Cement

additives can be used to:

- Vary the slurry density
- Change the compressive strength
- Accelerate or retard the setting time
- Control filtration and fluid loss
- Reduce slurry viscosity

Additives may be delivered to the rig in granular or liquid form and may be blended with the cement

powder or added to the mixwater before the slurry is mixed. The amount of additive used is usually given

in terms of a percentage by weight of the cement powder (based on each sack of cement weighing 94 lb).

Several additives will affect more than one property and so care must be taken as to how they are used

(Figure 4).

It should be remembered that the slurry is mixed up and tested in the laboratory before the actual cement

job.

(a) Accelerators

Accelerators are added to the cement slurry to shorten the time taken for the cement to set. These are used

when the setting time for the cement would be much longer than that required to mix and place the slurry,

and the drilling rig would incur WOC time. Accelerators are especially important in shallow wells where

temperatures are low and therefore the slurry may take a long time to set. In deeper wells the higher

temperatures promote the setting process, and accelerators may not be necessary.

The most common types of accelerator are:

- Calcium chloride (CaCl_2) 1.5 - 2.0%
- Sodium chloride (NaCl) 2.0 - 2.5%
- Seawater

It should be noted that at higher concentrations these additives will act as retarders.

(b) Retarders

In deep wells the higher temperatures will reduce the cement slurry's thickening time. Retarders are used

to prolong the thickening time and avoid the risk of the cement setting in the casing prematurely. The

bottom hole temperature is the critical factor which influences slurry setting times and therefore for determining the need for retarders. Above a static temperature of 260 - 275 degrees F the effect of retarders

should be measured in pilot tests.

The most common types of retarders are:

- Calcium lignosulphanate (sometimes with organic acids) 0.1 - 1.5%
- Saturated Salt Solutions

(c) Lightweight additives (Extenders)

Extenders are used to reduce slurry density for jobs where the hydrostatic head of the cement slurry may

exceed the fracture strength of certain formations. In reducing the slurry density the ultimate compressive

strength is also reduced and the thickening time increased. The use of these additives allows more

mixwater to be added, and hence increases the amount of slurry which is produced by each sack of cement

powder (the yield of the slurry). Such additives are therefore sometimes called extenders.

The most common types of lightweight additives are:

- Bentonite (2 - 16%) - This is by far the most common type of additive used to lower slurry density. The

bentonite material absorbs water, and therefore allows more mixwater to be added. Bentonite will also

however reduce compressive strength and sulphate resistance. The increased yield due to the bentonite

added is shown in Table

- Pozzolan - This may be used in a 50/50 mix with the Portland cement. The result is a slight decrease in

compressive strength, and increased sulphate resistance.

- Diatomaceous earth (10 - 40%) - The large surface area of diatomaceous earth allows more water absorption, and produces low density slurries (down to 11ppg).

(d) Heavyweight additives

Heavyweight additives are used when cementing through overpressured zones. The most common types

of additive are:

- Barite (barium sulphate) - this can be used to attain slurry densities of up to 18ppg. It also causes a

reduction in strength and pumpability.

- Hematite (Fe_2O_3) - The high specific gravity of hematite can be used to raise slurry densities to 22 ppg. Hematite significantly reduces the pumpability of slurries and therefore friction reducing additives may be required when using hematite.

- Sand - graded sand (40 - 60 mesh) can give a 2 ppg increase in slurry density.

(e) Fluid loss additives

Fluid loss additives are used to prevent dehydration of the cement slurry and premature setting. The most

common additives are:

- Organic polymers (cellulose) 0.5 - 1.5%

- Carboxymethyl hydroxyethyl cellulose (CMHEC) 0.3 - 1.0% (CMHEC will also act as a retarder)

(f) Friction reducing additives (Dispersants)

Dispersants are added to improve the flow properties of the slurry. In particular they will lower the viscosity of the slurry so that turbulence will occur at a lower circulating pressure, thereby reducing the

risk of breaking down formations.

The most commonly used are:

- Polymers 0.3 - 0.5 lb/sx of cement

- Salt 1 - 16 lb/sx

- Calcium lignosulphanate 0.5 - 1.5 lb/sxg)

(g) Mud contaminates

As well as the compounds deliberately added to the slurry on surface, to improve the slurry properties, the cement slurry will also come into contact with, and be contaminated by, drilling mud when it is pumped downhole. The chemicals in the mud may react with the cement to give undesirable side effects.

Some of these are listed below:

Mud additive Effect on cement

barite increases density and reduces compressive strength

caustic acts as an accelerator

calcium compounds decrease density

diesel oil decrease density

thinners act as retarders

The mixture of mud and cement causes a sharp increase in viscosity. The major effect of a highly viscous

fluid in the annulus is that it forms channels which are not easily displaced. These channels prevent a good

cement bond all round the casing.

To prevent mud contamination of the cement a spacer fluid is pumped ahead of the cement slurry.

4. PRIMARY CEMENTING

The objective of a primary cement job is to place the cement slurry in the annulus behind the casing. In

most cases this can be done in a single operation, by pumping cement down the casing, through the

casing shoe and up into the annulus. However, in longer casing strings and in particular where the formations are weak and may not be able to support the hydrostatic pressure generated by a very long

column of cement slurry, the cement job may be carried out in two stages. The first stage is completed in

the manner described above, with the exception that the cement slurry does not fill the entire annulus, but

reaches only a pre-determined height above the shoe. The second stage is carried out by including a

special tool in the casing string which can be opened, allowing cement to be pumped from the casing

and into the annulus. This tool is called a multi stage cementing tool and is placed in the casing string

at the point at which the bottom of the second stage is required. When the second stage slurry is ready to

be pumped the multi stage tool is opened and the second stage slurry is pumped down the casing, through

the stage cementing tool and into the annulus, as in the first stage. When the required amount of slurry has

been pumped, the multi stage tool is closed. This is known as a two stage cementing operation and will

be discussed in more detail later.

The height of the cement sheath, above the casing shoe, in the annulus depends on the particular objectives of the cementing operations. In the case of conductor and surface casing the whole annulus

is generally cemented so that the casing is prevented from buckling under the very high axial loads produced by the weight of the wellhead and BOP.

In the case of the intermediate and production casing the top of the cement sheath (Top of Cement - TOC)

is generally selected to be approximately 300-500 ft. above any formation that could cause problems in

the annulus of the casing string being cemented. For instance, formations that contain gas which could

migrate to surface in the annulus would be covered by the cement. Liners are generally cemented over

their entire length, all the way from the liner shoe to the liner hanger.

4.1 Downhole cementing equipment

In order to carry out a conventional primary cement job some special equipment must be included in the

casing string as it is run.

a. Guide shoe - A guide (Figure 5) shoe is run on the bottom of the first joint of casing. It has a rounded nose to guide the casing past any ledges or other irregularities in the hole.

b. Float collar - A float collar (Figure 6) is positioned 1 or 2 joints above the guide shoe. It acts as a seat for the cement plugs used in the pumping and displacement of the cement slurry. This means that at the end of the cement job there will be some cement left in the casing between the float collar and the guide shoe which must be drilled out.

The float collar also contains a non-return valve so that the cement slurry cannot flow back up the casing. This is necessary because the cement slurry in the annulus generally has a higher density than the

displacing fluid in the casing, therefore a U-tube effect is created when the cement is in position and the

pumps are stopped.

Sometimes the guide shoe also has a non-return valve as an extra precaution. It is essential that the non-

return valves are effective in holding back the cement slurry.

The use of a non-return valve means that as the casing is being run into the borehole the fluid in the hole

cannot enter the casing from below. This creates a buoyancy effect which can be reduced by filling up the

casing from the surface at regular intervals while the casing is being run (every 5 - 20 joints). This filling

up process increases the running in time and can be avoided by the use of automatic or differential fill up

devices fitted to the float collar or shoe. These devices allow a controlled amount of fluid to enter the

casing at the bottom of the string. The ports through which the fluid enters are blocked off before the

cement job begins. The use of a differential fill-up device also reduces the effect of surge pressures on the

formation.

c. Centralisers - these are hinged metal ribs which are installed on the casing string as it is run (Figure 7). Their function is to keep the casing away from the borehole so that there is some annular

clearance around the entire circumference of the casing

The proper use of centralisers will help to:

- Improve displacement efficiency (i.e. place cement all the way around the casing)
- Prevent differential sticking
- Keep casing out of keyseats

Centralisers are particularly required in deviated wells where the casing tends to lie

on the low side of the hole. On the high side there will be little resistance to flow, and so cement placement

will tend to flow up the high side annular space. Mud channels will tend to form on the low side of the

hole, preventing a good cement job. Each centraliser is hinged so that it can be easily clamped onto the

outside of the casing and secured by a retaining pin. The centraliser is prevented from moving up and

down the casing by positioning the centraliser across a casing coupling or a collar known as a stop collar.

The spacing of centralisers will vary depending on the requirements of each cement job. In critical zones,

and in highly deviated parts of the well, they are closely spaced, while on other parts of the casing string

they may not be necessary at all.

A typical programme might be:

1 centraliser immediately above the shoe

1 every joint on the bottom 3 joints

1 every joint through the production zone

1 every 3 joints elsewhere

d. Wipers/scratchers - these are devices run on the outside of the casing to remove mud cake and break up gelled mud. They are sometimes used through the production zone.

4.2 Surface cementing equipment

a. Mixing and pumping facilities:

On most rigs cement powder and additives are handled in bulk, which makes blending and mixing much easier. For large volume cement jobs several bulk storage bins may be required on the rig. On offshore rigs the cement is transferred pneumatically from supply boats to the storage bins. For any cement job there must be sufficient water available to mix the slurry at the desired water/cement ratio when required. The mix water must also be free of all contaminants. The water is added to the cement in a jet mixer (Figure 8). The mixer consists of a funnel shaped hopper, a mixing bowl, a water supply line and an outlet for the slurry. As the mixwater is pumped across the lower end of the hopper a venture effect is created and cement powder is drawn down into the flow of mixwater and a slurry is created. The slurry flows into a slurry tub where its density is measured. The density of the slurry should be regularly checked during the cement job since this is the primary means by which the quality of the slurry is determined. If the density of the slurry is correct then the correct amount of mixwater has been mixed with the cement powder. Samples can be taken directly from the mixer and weighed in a standard mud balance or automatic devices (densometers) can also be used. Various types of cement pumping units are available. For land based jobs they can be mounted on a truck, while skid mounted units are used offshore. The unit normally has twin pumps (triplex, positive displacement) which may be diesel powered or driven by electric motors. These units can operate at high pressures (up to 20,000 psi) but are generally limited to low pumping rates. Most units are capable of mixing and displacing 50 - 70 cubic feet of slurry per minute. In order to minimise contamination by the mud in the annulus a preflush or spacer fluid is pumped ahead of the cement slurry. The actual composition of the spacer depends on the type of mud being used. For water based muds the spacer fluid is often just water, but specially designed fluids are available. The volume of spacer is based on the need to provide sufficient separation of mud and cement in the annulus (20 - 50 bbls of spacer is common).

b. Cementing heads:

Cementing heads:

The cement head provides the connection between the discharge line from the cement unit and the top of the casing (Figure 9). This piece of equipment is designed to hold the cement plugs used in the

conventional primary cement job.

The cement head makes it possible to release the bottom plug, mix and pump down the cement slurry,

release the top plug and displace the cement without making or breaking the connection to the top of the

casing. For ease of operation the cement head should be installed as close to rig floor level as possible.

The cement jobs will be unsuccessful if the cement plugs are installed in the correct sequence or are not

released from the cementing head.

Mud is normally used to displace the cement slurry. The cement pumps or the rig pumps may be used for

the displacement. It is recommended that the cement slurry is displaced at as high a rate as possible. High

rate displacement will aid efficient mud displacement. It is highly unlikely that it will be possible to achieve turbulence

in the cement slurry since it is so viscous and has such a high density. However, it may be possible to

generate turbulence in the spacer and this will result in a more efficient displacement of the mud.

4.3 Single Stage Cementing Operation

The single stage primary cementing operation is the most common type of cementing operation that is conducted when drilling a well. The procedure for performing a single stage cementing

operation (Figure 10) will be discussed first and then the procedure for conducting a multiple stage and

stinger cementing operations will be discussed.

In the case of the single stage operation, the casing with all of the required cementing accessories such as

the float collar, centralisers etc. is run in the hole until the shoe is just a few feet off the bottom of the hole

and the casing head is connected to the top of the casing. It is essential that the cement plugs are correctly

placed in the cement head. The casing is then circulated clean before the cementing operation begins (at

least one casing volume should be circulated). The first cement plug (wiper plug) shown in Figure 11, is

pumped down ahead of the cement to wipe the inside of the casing clean. The spacer is then pumped into

the casing. The spacer is followed by the cement slurry and this is followed by the second plug (shut-off

plug) shown in Figure 12. When the wiper plug reaches the float collar its rubber diaphragm is ruptured,

allowing the cement slurry to flow through the plug, around the shoe, and up into the annulus. At this

stage the spacer is providing a barrier to mixing of the cement and mud. When the solid, shut-off plug reaches the float collar it lands on the wiper plug and stops the displacement process. The pumping rate should be slowed down as the shut-off plug approaches the float collar and the shut-off plug should be gently bumped into the bottom, wiper plug. The casing is often pressure tested at this point in the operation. The pressure is then bled off slowly to ensure that the float valves, in the float collar and/or casing shoe, are holding.

The displacement of the top plug is closely monitored. The volume of displacing fluid necessary to bump the plug should be calculated before the job begins. When the pre-determined volume has almost been completely pumped, the pumps should be slowed down to avoid excessive pressure when the plug is bumped. If the top plug does not bump at the calculated volume (allowing for compression of the mud) this may be because the top, shut-off plug has not been released. If this is the case, no more fluid should be pumped, since this would displace the cement around the casing shoe and up the annulus.

Throughout the cement job the mud returns from the annulus should be monitored to ensure that the formation has not been broken down. If formation breakdown does occur then mud returns would slow down or stop during the displacement operation.

The single stage procedure can be summarised as follows:

1. Circulate the casing and annulus clean with mud (one casing volume pumped)
2. Release wiper (top) plug
3. Pump spacer
4. Pump cement
5. Release shut-off (bottom) plug
6. Displace with displacing fluid (generally mud) until the shut-off plug lands on the float collar
7. Pressure test the casing

4.4 Multi - Stage Cementing Operation

When a long intermediate string of casing is to be cemented it is sometimes necessary to split the cement sheath in the annulus into two, with one sheath extending from the casing shoe to some point above potentially troublesome formations at the bottom of the hole, and the second sheath covering shallower troublesome formations. The placement of these cement sheaths is known as a multi-stage cementing

operation (Figure 13).

The reasons for using a multi-stage operation are to reduce:

- Long pumping times
- High pump pressures
- Excessive hydrostatic pressure on weak formations due to the relatively high density of cement slurries.

The procedure for conducting a multi-stage operation is as follows:

First stage

The procedure for the first stage of the operation is similar to that described in Section 4.3 above, except

that a wiper plug is not used and only a liquid spacer is pumped ahead of the cement slurry. The conventional shut-off plug is replaced by a plug with flexible blades. This type of shut-off plug is used

because it has to pass through the stage cementing collar which will be discussed below. It is worth noting

that a smaller volume of cement slurry is used, since only the lower part of the annulus is to be cemented.

The height of this cemented part of the annulus will depend on the fracture gradient of the formations

which are exposed in the annulus (a height of 3000' - 4000' above the shoe is common).

Second stage

The second stage of the operation involves the use of a special tool known as a stage collar (Figure 14),

which is made up into the casing string at a pre-determined position. The position often corresponds to

the depth of the previous casing shoe.

The ports in the stage collar are initially sealed off by the inner sleeve. This sleeve is held in place by

retaining pins. After the first stage is complete a special dart is released from surface which lands in the

inner sleeve of the stage collar. When a pressure of 1000 - 1500 psi is applied to the casing above the dart,

and therefore to the dart, the retaining pins on the inner sleeve are sheared and the sleeve moves down,

uncovering the ports in the outer mandrel. Circulation is established through the stage collar before the

second stage slurry is pumped.

The normal procedure for the second stage of a two stage operation is as follows:

- 1 Drop opening dart
- 2 Pressure up to shear pins
- 3 Circulate through stage collar whilst the first stage cement is setting
- 4 Pump spacer
- 5 Pump second stage slurry

6 Release closing plug

7 Displace plug and cement with mud

8 Pressure up on plug to close ports in stage collar and pressure test the casing.

To prevent cement falling down the annulus a cement basket or packer may be run on the casing below

the stage collar. If necessary, more than one stage collar can be run on the casing so that various sections

of the annulus can be cemented. One disadvantage of stage cementing is that the casing cannot be moved

after the first stage cement has set in the lower part of the annulus. This increases the risk of channelling

and a poor cement bond.

4.5 Inner string cementing

For large diameter casing, such as conductors and surface casing, conventional cementing techniques

result in:

- The potential for cement contamination during pumping and displacement
- The use of large cement plugs which can get stuck in the casing
- Large displacement volumes
- Long pumping times
- Large volume of cement left inside the casing between float collar and shoe.

An alternative technique, known as a stinger cement job, is to cement the casing through a tubing or

drillpipe string, known as a cement stinger, rather than through the casing itself.

In the case of a stinger cement job the casing is run as before, but with a special float shoe (Figure 15)

rather than the conventional shoe and float collar. A special sealing adapter, which can seal in the seal

bore of the seal float shoe, is attached to the cement stinger. Once the casing has been run, the cementing

string (generally tubing or drillpipe), with the seal adapter attached, is run and stabbed into the float shoe.

Drilling mud is then circulated around the system to ensure that the stinger and annulus are clear of any

debris. The cement slurry is then pumped with liquid spacers ahead and behind the cement slurry. No

plugs are used in this type of cementing operation since the diameter of the stinger is generally so small

that contamination of the cement is unlikely if a large enough liquid spacer is used.

The cement slurry is generally under-displaced so that when the seal adapter on the stinger is pulled from

the shoe the excess cement falls down on top of the shoe. This can be subsequently drilled out when the

next hole section is being drilled. Under-displacement however ensures that the cement slurry is not

displaced up above the casing shoe, leaving spacer and drilling mud across the shoe. After the cement has

been displaced, and the float has been checked for backflow, the cement stinger can be retrieved. This

method is suitable for casing diameters of 13 3/8" and larger. The main disadvantage of this method is

that for long casing strings rig time is lost in running and retrieving the inner string.

4.6 Liner cementing

Liners are run on drillpipe and therefore the conventional cementing techniques cannot be used for cementing a liner. Special equipment must be used for cementing these liners.

As with a full string of casing the liner has a float collar and shoe installed. In addition there is a landing

collar, positioned about two joints above the float collar (Figure 16). A wiper plug is held on the end of

the tailpipe of the running string by shear pins.

The liner is run on drillpipe and the hanger is set at the correct point inside the previous casing string.

Mud is circulated to ensure that the liner and the annulus is free from debris, and to condition the mud.

Before the cementing operation begins the liner setting tool is backed off to ensure that it can be recovered

at the end of the

cement job.

The cementing procedure is as follows:

1 Pump spacer ahead of cement slurry

2 Pump slurry

3 Release pump down plug

4 Displace cement down the running string and out of the liner into the annulus

5 Continue pumping until the pump down plug lands on the wiper plug.

6 Apply pressure to the pump down plug and shear out the pins on the wiper plug. This releases the wiper

plug

7 Both plugs move down the liner until they latch onto landing collar

8 Bump the plugs with 1000 psi pressure

9 Bleed off pressure and check for back flow

Since there is no bottom plug in front of the slurry the wiper plug cleans off debris and mud from the

inside of the liner. This material will contaminate the cement immediately ahead of the wiper plug. The

spacing between the landing collar and the shoe should be adequate to accommodate this contaminated

cement, and thus prevent it from reaching the annulus where it would create a poor cement job around the shoe.

To promote a good cement job, cement in excess of that required to fill the annulus between the liner and

the borehole is used. This excess cement will pass up around the liner top and settle on top of the liner

running assembly. Once the cement is in place the liner setting tool is quickly picked up out of the liner.

With the tail pipe above the liner top the excess cement can be reverse circulated out. The setting tool can

then be retrieved.

In practice it is very difficult to obtain a good cement job on a liner.

The main reasons for this are:

(a) Minimal annular clearances

A 7" OD liner run in an 8 1/2" hole gives a clearance of only 3/4" (assuming the liner is perfectly centred).

This small clearance means that:

- It is difficult to run the liner (surge pressure)
- High pressure drops occur during circulation (lost circulation problems)
- It is difficult to centralise the liner
- Cement placement is poor (channeling)

(b) Mud contamination

When the cement comes in contact with mud or mud cake it may develop high viscosity. The increased

pump pressure required to move this contaminated cement up the annulus may cause formation breakdown. Fluid loss additives must be used to prevent dehydration of the cement which may cause

bridging in the annulus.

(c) Lack of pipe movement

Due to risk of sticking the setting tool, most operators want to be free of the liner before cementing begins.

By disconnecting the setting tool the liner cannot be moved during the cement job. This lack of movement

reduces the efficiency of cement placement. Due to these problems it is often necessary to carry out a

remedial squeeze job at the top of the liner (Figure17). It is becoming more common these days to remain

latched on top of the liner and rotate the liner whilst the cement is being displaced into position. A special

piece of liner running equipment, known as a rotating liner assembly, is used for this purpose

4.7 Recommendations for a good cement job

The main cause for poor isolation after a cement job is the presence of mud channels in the cement sheath

in the annulus. These channels of gelled mud exist because the mud in the annulus has not been displaced

by the cement slurry. This can occur for many reasons. The main reason for this is poor centralisation of

the casing in the borehole, during the cementing operation. When mud is being displaced from the annulus

the cement will follow the least path of resistance. If the pipe is not properly centralised the highest resistance to flow occurs where the clearance is least. This is where mud channels are most likely to occur

(Figure 18).

In addition, field tests have shown that for a good cement bond to develop the formation should be in

contact with the cement slurry for a certain time period while the cement is being displaced. The recommended contact time (pump past time) is about 10 minutes for most cement jobs. To improve mud

displacement and obtain a good cement bond the following practices are recommended:

- Use centralisers, especially at critical points in the casing string
- Move the casing during the cement job. In general, rotation is preferred to reciprocation, since the latter

may cause surging against the formation. A specially designed swivel may be installed between the cementing head and the casing to allow rotation. (Centralisers remain static and allow the casing to rotate

within them.)

- Before doing the cement job, condition the mud (low PV, low YP) to ensure good flow properties, so

that it can be easily displaced.

- Displace the spacer in turbulent flow. This may not be practicable in large diameter casing where the

high pump rates and pressures may cause erosion or formation breakdown.

- Use spacers to prevent mud contamination in the annulus

5. SQUEEZE CEMENTING

Squeeze cementing is the process by which hydraulic pressure is used to force cement slurry through holes

in the casing and into the annulus and/or the formation.

Squeeze cement jobs are often used to carry out remedial operations during a workover on the well (Figure

3).

The main applications of squeeze cementing are:

- To seal off gas or water producing zones, and thus maximise oil production from the completion interval
- To repair casing failures by squeezing cement through leaking joints or corrosion hole
- To seal off lost circulation zones

- To carry out remedial work on a poor primary cement job (e.g. to fill up the annulus)
- To prevent vertical reservoir fluid migration into producing zones (block squeeze)
- To prevent fluids escaping from abandoned zones.

During squeeze cementing the pores in the rock rarely allow whole cement to enter the formation since a

permeability of about 500 darcies would be required for this to happen. There are two processes by which

cement can be squeezed:

- High pressure squeeze - This technique requires that the formation be fractured. which then allows the cement slurry to be pumped into the fractured zone.

- Low pressure squeeze - During this technique the fracture gradient of the formation is not exceeded.

Cement slurry is placed against the formation, and when pressure is applied the fluid content (filtrate) of

the cement is squeezed into the rock, while the solid cement material (filter cake) builds up on the face of

the formation.

5.1 High Pressure Squeeze

In a high pressure squeeze the formation is initially fractured (broken down) by a solids free breakdown

fluid. A solids free fluid is used because a solids laden fluid such as drilling mud will build up a filter cake and prevent injection into the formation. Solids free fluids such as water or brine are recommended.

The direction of the fracture depends on the rock stresses present in the formation. The fracture will occur

along a plane perpendicular to the direction of the least compressive stress (Figure 19). In general, the

vertical stress, due to the overburden, will be greater than the horizontal stresses. A vertical fracture is

therefore more likely.

In practice the fracture direction is difficult to predict since it may follow natural fractures in the formation.

Since squeeze cementing is often used to isolate various horizontal zones a vertical fracture is of little use

(vertical fluid movement is not prevented)

After the formation is broken down a slurry of cement is spotted adjacent to the formation, and then

pumped into the zone at a slow rate. The injection pressure should gradually build up as the cement fills

up the fractured zone. After the cement has been squeezed the pressure is released to check for back flow.

The disadvantages of this technique are:

- No control over the orientation of the fracture

- Large volumes of cement may be necessary to seal off the fracture
- Mud filled perforations may not be opened up by fracturing, so the cement may not seal them off effectively.

5.2 Low Pressure Squeeze

It is generally accepted that a low pressure squeeze is a more efficient method of sealing off unwanted

perforated zones. In a low pressure squeeze the formation is not fractured. Instead a cement slurry is gently

squeezed against the formation. A cement slurry consists of finely divided solids dispersed in a liquid. The

solids are too large to be displaced into the formation. As pressure is applied, the liquid phase is forced

into the pores, leaving a deposit of solid material or filter cake behind. As the filter cake of dehydrated

cement begins to build up, the impermeable barrier prevents further filtrate invasion. The filtrate must

then be diverted to other parts of the perforated interval. This technique therefore creates an impermeable

seal across the perforated zone. Fluid loss additives are important to perform this technique successfully.

Neat cement has a high fluid loss, resulting in rapid dehydration which causes bridging before the other

perforations are sealed off. Conversely a very low fluid loss means a slow filter cake build up and long

cement placement time.

Key factors which affect the build up of cement filter cake are:

- Fluid loss (generally 50 - 200 cc)
- Water to solids ratio (0.4 by weight)
- Formation characteristics (permeability, pore pressure)
- Squeeze pressure

Only a small volume of cement is required for a low pressure squeeze. Perforations must be free from

mud or other plugging material. If the well has been producing for some time these perforations have to

be washed out, sometimes with an acid solution. The general procedure for a low pressure squeeze job is:

1 Water is pumped into the zone to establish whether the formation can be squeezed (injectivity test). If

water cannot be injected the squeeze job cannot be done without fracturing the formation

2 Spot the cement slurry at the required depth

3 Apply moderate squeeze pressure

4 Stop pumping and check for bleed off

5 Continue pumping until bleed off ceases for about 30 mins

6 Stop displacement of cement and hold pressure

7 Reverse circulate out excess cement from casing

A properly designed slurry will leave only a small cement node inside the casing after removing the excess

cement. Throughout the procedure squeeze pressure is kept below the fracture gradient. A running squeeze is where the cement is pumped slowly and continuously until the final squeeze pressure is obtained. This is often used for repairing a primary cement job. A hesitation squeeze is where pumping

is stopped at regular intervals to allow time for the slurry to dehydrate and form a filter cake. Small volumes of cement (1/4 - 1/2 bbl) are pumped each time separated by a delay of 10 - 15 mins. This technique is dangerous if the cement is still in contact with the drillpipe or packer.

5.3 Equipment Used for Squeeze Cementing

The high pressure and low pressure squeeze operations can be conducted with or without packers.

(a) Bradenhead squeeze

This technique involves pumping cement through drill pipe without the use of a packer (Figure 20). The

cement is spotted at the required depth. The BOPs and the annulus are closed in and displacing fluid is

pumped down, forcing the cement into the perforations, since it cannot move up the annulus. This is the

simplest method of placing and squeezing cement, but has certain disadvantages:

- It is difficult to place the cement accurately against the target zone
- It cannot be used for squeezing off one set of perforations if other perforations are to remain open
- Casing is pressured up, and so squeeze pressure is limited by burst resistance

A Bradenhead squeeze is only generally used for a low-pressure squeeze job.

(b) Squeeze using a packer

The use of a packer makes it possible to place the cement more accurately, and apply higher squeeze

pressures. The packer seals off the annulus, but allows communication between drill pipe and the wellbore

beneath the packer. (Figure 21)

Two types of packer may be used in this type of operation:

(i) Drillable packer (cement retainer)

This type of packer contains a back pressure valve which will prevent the cement flowing back after the

squeeze. These are mainly used for remedial work on primary cement jobs, or to close off water producing

zones. The packer is run on drill pipe or wireline and set just above or between sets of perforations. When

the cement has been squeezed successfully the drill pipe can be removed, closing the back pressure valve.

The advantages of these packers are:

- Good depth control

- Back pressure valve prevents cement back flowing
- Drill pipe recovered without disturbing cement

The major disadvantage is that they can only be used once then drilled out.

(ii) Retrievable packer (cement retainer)

These can be set and released many times on one trip. This makes them suitable for repairing a series of

casing leaks or selectively squeezing off sets of perforations.

By-pass ports in the packer allow annular communication, but these ports are closed during the squeeze

job. When the packer is released there may be some backflow, and the cement filter cake may be disturbed.

If this happens the packer should be re-set and the squeeze pressure applied until the cement sets.

The basic procedure for squeezing with a retrieveable packer is:

1. run the packer on drillpipe and set it at required depth with by-pass open
2. pump the cement slurry (keep back pressure on annulus to prevent cement falling

The packer setting depth should be considered carefully. If positioned too high above the perforations the

slurry will be contaminated by the wellbore fluids and large volumes of fluid from below the packer will

be pumped into the formation ahead of the cement. If the packer is set too low it may become stuck in the

cement. Generally the packer is set 30 - 50 ft above the perforations.

Sometimes a tail pipe is used below the packer to ensure that only cement is squeezed into the perforations,

and there is less chance of getting stuck (Figure 21). Bridge plugs are often set in the wellbore, to isolate

zones which are not to be treated.

They seal off the entire wellbore, and hold pressure from above and below. Bridge plugs can either be

drillable or retrievable.

5.4 Testing the Squeeze Job

After the cement has hardened it must be pressure tested. The tests should include both positive and

negative differential pressure. The following should be considered when making a test:

- A positive pressure test can be performed by closing the BOPs and pressuring up on the casing. (Do not

exceed formation fracture gradient.)

- A negative pressure test (or inflow test) can be performed by reducing the hydrostatic pressure inside

the casing. This can be done using a DST tool or displacing with the well to diesel. This test is more meaningful since mud filled perforations may hold pressure from the casing, but may become unblocked

when pressure from the formation is applied.

6. CEMENT PLUGS

At some stage during the life of a well a cement plug may have to be placed in the wellbore. A cement

plug is designed to fill a length of casing or open hole to prevent vertical fluid movement.

Cement plugs may be used for:

- Abandoning depleted zones
- Seal off lost circulation zones
- Providing a kick off point for directional drilling (eg side- tracking around fish)
- Isolating a zone for formation testing
- Abandoning an entire well (government regulations usually insist on leaving a series of cement plugs in

the well prior to moving off location).

The major problem when setting cement plugs is avoiding mud contamination during placement of the cement. Certain precautions should be taken to reduce contamination.

- Select a section of clean hole which is in gauge, and calculate the volume required (add on a certain amount of excess). The plug should be long enough to allow for some contamination (500' plugs are

common). The top of the plug should be 250' above the productive zone

- Condition the mud prior to placing the plug
- Use a preflush fluid ahead of the cement
- Use densified cement slurry (ie less mixwater than normal)

After the cement has hardened the final position of the plug should be checked by running in and tagging

the cement. There are three commonly used techniques for placing a cement plug:

(a) Balanced plug (Figure 22)

This method is aimed at achieving an equal level of cement in the drillpipe and annulus. Preflush, cement

slurry and spacer fluid are pumped down the drillpipe and displaced with mud. The displacement continues

until the level of cement inside and outside the drillpipe is the same (hence balanced). If the levels are not

the same then a U-tube effect will take place. The drillpipe can then be retrieved leaving the plug in place.

(b) Dump bailer (Figure 23)

A dump bailer is an electrically operated device which is run on wireline. A permanent bridge plug is set

below the required plug back depth. A cement bailer containing the slurry is then lowered down the well

on wireline. When the bailer reaches the bridge plug the slurry is released and sits on top of the bridge

plug.

The advantages of this method are:

- High accuracy of depth control
- Reduced risk of contamination of the cement the disadvantages are:
- Only a small volume of cement can be dumped at a time - several runs may be necessary
- It is not suitable for deep wells, unless retarders used.

7. EVALUATION OF CEMENT JOBS

A primary cement job can be considered a failure if the cement does not isolate undesirable zones. This

will occur if:

- The cement does not fill the annulus to the required height between the casing and the borehole.
 - The cement does not provide a good seal between the casing and borehole and fluids leak through the cement sheath to surface.
 - The cement does not provide a good seal at the casing shoe and a poor leak off test is achieved
- When

any such failures occur some remedial work must be carried out.

A number of methods can be used to assess the effectiveness of the cement job. These include:

Detecting Top of Cement (TOC)

(a) Temperature surveys (Figure 24)

This involves running a thermometer inside the casing just after the cement job.

The thermometer responds to the heat generated by the cement hydration, and so can be used to detect the

top of the cement column in the annulus.

(b) Radioactive surveys (Figure 25)

Radioactive tracers can be added to the cement slurry before it is pumped (Carnolite is commonly used).

A logging tool is then run when the cement job is complete.

This tool detects the top of the cement in the annulus, by identifying where the radioactivity decreases to

the background natural radioactivity of the formation.

Detecting Top of Cement (TOC) and the Measuring the Quality of the Cement Bond

(a) Cement bond logs (CBL)

The cement bond logging tools have become the standard method of evaluating cement jobs since they

not only detect the top of cement, but also indicate how good the cement bond is. The CBL tool is basically

a sonic tool which is run on wireline. The distance between transmitter and receiver is about 3 ft (Figure

26). The logging tool must be centralised in the hole to give accurate results. Both the time taken for the

signal to reach the receiver, and the amplitude of the returning signal, give an indication of the cement

bond. Since the speed of sound is greater in casing than in the formation or mud the first signals which

are received at the receiver are those which travelled through the casing (Figure 27). If the amplitude (E1)

is large (strong signal) this indicates that the pipe is free (poor bond). When cement is firmly bonded to

the casing and the formation the signal is attenuated, and is characteristic of the formation behind the

casing.

(b) the Variable Density Log (VDL)

The CBL log usually gives an amplitude curve and provides an indication of the quality of the bond between the casing and cement. A VDL (variable density log), provides the wavetrain of the received

signal (Figure 28), and can indicate the quality of the cement bond between the casing and cement, and

the cement and the formation. The signals which pass directly through the casing show up as parallel,

straight lines to the left of the VDL plot. A good bond between the casing and cement and cement

formation is shown by wavy lines to the right of the VDL plot. The wavy lines correspond to those signals

which have passed into and through the formation before passing back through the cement sheath and

casing to the receiver. If the bonding is poor the signals will not reach the formation and parallel lines will

be recorded all across the VDL plot.

The interpretation of CBL logs is still controversial. There is no standard API scale to measure the effectiveness of the cement bond. There are many factors which can lead to false interpretation:

- During the setting process the velocity and amplitude of the signals varies significantly. It is recommended that the CBL log is not run until 24 - 36 hours after the cement job to give realistic results.

- Cement composition affects signal transmission

- The thickness of the cement sheath will cause changes in the attenuation of the signal

- The CBL will react to the presence of a microannulus (a small gap between casing and cement).

The

microannulus usually heals with time and is not a critical factor. Some operators recommend running the

CBL under pressure to eliminate the microannulus effect

Cementing Calculations

The following calculations must be undertaken prior to a cementation operation:

- Slurry Requirements

- No. of sacks of Cement

- Volume of Mixwater

- Volume of Additives

- Displacement Volume Duration of Operation

These calculations will form the basis of the cementing programme. They should be performed in this

sequence as will be seen below.

1. Cement Slurry Requirements :

Sufficient cement slurry must be mixed and pumped to fill up the following (see Fig 29):

A - the annular space between the casing and the borehole wall,

B - the annular space between the casings (in the case of a two stage cementation operation)

C - the openhole below the casing (rathole)

D - the shoetrack

The volume of slurry that is required will dictate the amount of dry cement, mixwater and additives that

will be required for the operation.

In addition to the calculated volumes an excess of slurry will generally be mixed and pumped to accommodate any errors in the calculated volumes. These errors may arise due to inaccuracies in the size

of the borehole (due to washouts etc.).

It is common to mix an extra 10-20% of the calculated openhole volumes to accommodate these inaccuracies.

The volumetric capacities (quoted in bbls/linear ft or cuft/linear ft or m³/m) of the annuli, casings, and

open hole are available from service company cementing tables.. These volumetric capacities can be

calculated directly but the cementing tables are simple to use and include a more accurate assessment of

the displacement of the casing for instance and the capacities based on nominal diameters.

In the case of a two stage operation (Figure 30) the volume of slurry used in the first stage of the operation

is the same as that for a single stage operation. The second stage slurry volume is the slurry required to

fill the annulus between the casing and hole (or casing/casing if the multi-stage collar is inside the previous

shoe) annular space.

2. Number of Sacks of Cement

Although cement and other dry chemicals are delivered to the rigsite in bulk tanks the amount of dry

cement powder is generally quoted in terms of the number of sacks (sxs) of cement required. Each sack

of cement is equivalent to 1 cu. ft of cement.

The number of sacks of cement required for the cement operation will depend on the amount of slurry

required for the operation (calculated above) and the amount of cement slurry that can be produced from

a sack of cement. The amount of cement slurry that can be produced from a sack of cement, known as the yield of the cement, will depend on the type of cement powder (API classification) and the amount of mixwater mixed with the cement powder. The latter will also depend on the type of cement and will vary with pressure and temperature. The number of sacks of cement required for the operation can be calculated from the following

$$\text{No. of Sacks} = \text{Total Volume of Slurry} \times \text{Yield of Cement}$$

3. Mixwater Requirements

The mixwater required to hydrate the cement powder will be prepared and stored in specially cleaned mud tanks. The amount of mixwater required for the operation will depend on the type of cement powder used.

The volume of mixwater required for the cement slurry can be calculated from:

$$\text{Mixwater Vol.} = \text{Mixwater per sack} \times \text{No. sacks}$$

4. Additive Requirements

There are a variety of additives which may be added to cement. These additives may be delivered to the rigsite as liquid or dry additives. The amount of additive is generally quoted as a percentage of the cement powder used. Since each sack of cement weighs 94 lbs, the amount of additive can be quoted in weight (lbs) rather than volume. This can then be related to the number of sacks of additive. The number of sacks of additive can be calculated from:

$$\text{Number of sacks of additive} = \text{No. sacks Cement} \times \% \text{ Additive}$$

$$\text{Weight of additive} = \text{No. sacks of Additive} \times 94(\text{lb/sk}).$$

The amount of additive is always based on the volume of cement to be used.

5. Displacement Volume

The volume of mud used to displace the cement from the cement stinger or the casing during the cementing operation is commonly known as the displacement volume. The displacement volume is dependant on the way in which the operation is conducted.

a. Stinger Operation :

The displacement volume can be calculated from the volumetric capacity of the cement stinger and the

depth of the casing shoe. The cement is generally under displaced by 1-2 bbls of liquid.

$$\text{Displacement Vol.} = \text{Volumetric capacity of stinger} \times \text{Depth of Casing} - 1\text{bbl}$$

b. Conventional Operation :

In a conventional cementing operation the displacement volume is calculated from the volumetric capacity

of the casing and the depth of the float collar in the casing.

Displacement Vol. = Volumetric Capacity of Casing x Depth of Float Collar

c. Two-stage Cementing Operation:

In a two stage operation the first stage is firstly displaced by a volume of mud, calculated in the same way

as a single stage cement operation described above.

The second stage displacement is then calculated on the basis of the volumetric capacity of the casing and

the depth of the second stage collar.

1st Stage :

Displacement Vol. = Volumetric Capacity of Casing x Depth of Float Collar

2nd stage :

Displacement Vol. = Volumetric Capacity of Casing x Depth of Multistage collar

The amount of mud to be pumped during the displacement operation may be quoted in terms of a volume

(bbbls, cuft etc.) or in terms of the number of strokes of the mud pump required to pump the mud volume.

It will therefore be necessary to determine the volume of fluid pumped with each stroke of the pumps

(vol./stroke). The number of strokes required to displace the cement will therefore be calculated from:

Number of strokes = Volume of displacement fluid/Vol. of fluid per stroke

6. Duration of Operation

The duration of the operation will be used to determine the required setting time for the cement formulation. The duration of the operation will be calculated on the basis of the mixing rate for the cement,

the pumping rate for the cement slurry and the pumping rate for the displacing mud. An additional period

of time, known as a contingency time, is added to the calculated duration to account for any operational

problems during the operation. This contingency is generally 1 hour in duration.

The duration of the operation can be calculated from:

Duration = Vol. of Slurry + Vol. of Slurry + Displacement Vol. + Contingency Time (1hr.)

Mixing Rate Pumping Rate Displacement Rate

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Chapter 1: Formation Pressure

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Well Design – Spring 2013

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Required Materials

1. Applied Drilling Engineering – Adam T. Bourgoyne – SPE

Textbook

2. Fundamental of Drilling Engineering – Miska and Mitchell – SPE

Textbook Volume 12

3. Drilling Engineering Handbook – Volume II – Robert Mitchell

4. Class notes

5. PowerPoint slides

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Grading

Homework 20%

Quizzes 25%

Project 20%

Final exam 35%

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Formation Pressure

Definition – Normal Pressure

During a period of erosion and sedimentation, grains of sediment are continuously building up on top of each other, generally in a water filled environment. As the thickness of the layer of sediment increases, the grains of the sediment are packed closer together, and some of the water is expelled from the pore spaces. However, if the pore throats through the sediment are interconnecting all the way to surface the

pressure of the fluid at any depth in the sediment will be same as that which would be found in a simple column of fluid. This pressure is called NORMAL PRESSURE and only depends on the density of the fluid in the pore space and the depth of the pressure measurement (equal to the height of the column of liquid). It will be independent of the pore size or pore throat geometry.

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Overburden Pressure

The vertical pressure at any point in the earth is known as the overburden pressure or geostatic pressure. The overburden pressure at any point is a function of the mass of rock and fluid above the point of interest. In order to calculate the overburden pressure at any point, the average density of the material (rock and fluids) above the point of interest must be determined. The average density of the rock and fluid in the pore space is known as the bulk density of the rock.

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Overburden Pressure

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Formation Pressure

Definition – Normal Pressure

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Formation Pressure

Definition – Normal Pressure

The datum which is generally used during drilling operations is the drillfloor elevation but a more general datum level, used almost universally, is Mean Sea Level, MSL. When the pore throats through the sediment are interconnecting, the pressure of the fluid at any depth in the sediment will be same as that which would be found in a simple column of fluid and therefore the pore pressure gradient is a straight line. The gradient of the line is a representation of the density of the fluid. Hence the density of the fluid in the pore space is often expressed in units of psi/ft.

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Formation Pressure

Definition – Abnormal Pressure

Pore pressures which are found to lie above or below the “normal” pore pressure

gradient line are called abnormal pore pressures. These formation pressures may be either Subnormal (i.e. less than 0.465 psi/ft) or Overpressured (i.e. greater than 0.465 psi/ft). The mechanisms which generate these abnormal pore pressures can be quite complex and vary from region to region. However, the most common mechanism for generating overpressures is called Undercompaction and can be best described by the undercompaction model.

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Formation Pressure

Definition – Abnormal Pressure

Underpressured

formation

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Abnormal Formation Pressure

Compact Effect

Prepared by: Tan Nguyen

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Abnormal Formation Pressure

Compact Effect

$P_{ob} = P_z - P_f$

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Causes of Abnormal Pressure

Subnormal Formation Pressure

(a) Formation Foreshortening

During a compression process there is some bending of strata. The upper beds can bend upwards, while the lower beds can bend downwards. The intermediate beds must expand to fill the void and so create a subnormally pressured zone. This is thought to apply to some subnormal zones in Indonesia and the US. Notice that this may also cause overpressures in the top and bottom beds.

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Causes of Abnormal Pressure

Subnormal Formation Pressure

(b) Thermal Expansion

As sediments and pore fluids are buried the temperature rises. If the fluid is allowed to expand the density will decrease, and the pressure will reduce.

(c) Depletion

When hydrocarbons or water are produced from a competent formation in which no subsidence occurs a subnormally pressured zone may result. This will be important when drilling development wells through a reservoir which has already been producing for some time. Some pressure gradients in Texas aquifers have been as low as 0.36 psi/ft.

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Causes of Abnormal Pressure

Subnormal Formation Pressure

(d) Potentiometric Surface: This mechanism refers to the structural relief of a formation and can result in both subnormal and overpressured zones. The potentiometric surface is defined by the height to which confined water will rise in wells drilled into the same aquifer. The potentiometric surface can therefore be thousands of feet above or below ground level

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Causes of Abnormal Pressure

Overpressured Formation

(a) Incomplete sediment compaction or undercompaction:

is the most common mechanism causing overpressures. In the rapid burial of low permeability clays or shales there is little time for fluids to escape. The formation pressure will build up and becomes overpressured formation. In other words, if the burial is rapid and the sand is enclosed by impermeable barriers, there is no time for this process to take place, and the trapped fluid will help to support the overburden.

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Causes of Abnormal Pressure

Overpressured Formation

(b) Faulting

Faults may redistribute sediments, and place permeable zones opposite impermeable zones, thus creating barriers to fluid movement. This may prevent water being expelled from a shale, which will cause high porosity and pressure within that shale under compaction.

(c) Massive Rock Salt Deposition

Deposition of salt can occur over wide areas. Since salt is impermeable to fluids, the underlying formations become overpressured. Abnormal pressures are frequently found in zones directly below a salt layer.

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Causes of Abnormal Pressure

Overpressured Formation

(d) Phase Changes during Compaction

Minerals may change phase under increasing pressure, e.g. gypsum ($\text{CaSO}_4 \cdot \text{H}_2\text{O}$) converts to anhydrite plus free water. It has been estimated that a phase change in gypsum will result in the release of water. The volume of water released is approximately 40% of the volume of the gypsum. If the water cannot escape then overpressures will be generated. Conversely, when anhydrite is hydrated at depth it will yield gypsum and result in a 40% increase in rock volume. The transformation of montmorillonite to illite also releases large amounts of water.

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Causes of Abnormal Pressure

Overpressured Formation

(e) Repressuring from Deeper Levels

This is caused by the migration of fluid from a high to a low pressure zone at shallower depth. This may be due to faulting or from a poor casing/cement job. The unexpectedly high pressure could cause a kick, since no lithology change would be apparent. High pressures can occur in shallow sands if they are charged by gas from lower formations.

(f) Generation of Hydrocarbons

Shales which are deposited with a large content of organic material will produce gas as the organic material degrades under compaction. If it is not allowed to escape the gas will cause overpressures to develop. The organic by-products will also form salts which will be precipitated in the pore space, thus helping to reduce porosity and create a seal.

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Compact Effect

Vertical overburden stress resulting from geostatic load at a sediment depth D:

D

□ ob □ □ □b gdD

0

□ b □ □ | □ □ □ g □ 1 □ □ □ ln □ □ ln □ o □ KDS

□ g □ □b 1 1

□□ Ds □ In □0 □ In □

$\square g \square \square f K K$

ϕ_0

In

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

K

D_s

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the depth below the surface of the sediments.

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Compact Effect

In offshore areas

D

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

D

D

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

D_s

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

D

D_s

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

D

D_s

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

D

D_s

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

K

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

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Compact Effect

Let D_s is the depth below the subsurface of the sediment.

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

K

ϕ_0 is the surface porosity, K is the porosity decline constant and D_s is the

$\rho_{ob} = \rho_{SW} g_{DSW} = g_{DS} = g_{\text{grain}} = g_{\text{fluid}}$
 $\rho_{1} = KDS = KDSW$
 K
 $e = e = KDSW$

$\rho_{ob} = \rho_{SW} g_{DSW} = g_{DS} = g_{\text{grain}} = g_{\text{fluid}}$
 $\rho_{1} = KDSW$
 K
 $e = e = KDS$

$g_{\text{grain}} = g_{\text{fluid}}$
 $\rho_{ob} = \rho_{SW} g_{DSW} = g_{DS}$
 K
 $\rho_{1} = e = KDS$

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Compact Effect

Example 1: Determine values for surface porosity and porosity decline constant K for the U.S. gulf coast area. Use the average grain density of 2.6 g/cm³, and average pore fluid density of 1.074 g/cm³.

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Compact Effect

$\ln$

0 -0.5 -1 -1.5 -2 -2.5 -3

0

2000 1 1

$Ds = \ln 0 = \ln$

4000 $K K$

6000

$y = -11681x - 10521$

8000

Ds, ft

10000

12000

14000

16000

18000

20000

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Compact Effect

$1/K = 11681 \text{ ft}$

$K = 8.56091 \times 10^{-5} \text{ ft}^{-1}$

$(1/K) \ln \phi_0 = -10521$

$\ln \phi_0 = -0.900693434$

$\phi_0 = 0.4087648$

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Compact Effect

Example 2:

Compute the vertical overburden stress resulting from geostatic load near the Gulf of Mexico coastline at a depth of 10,000 ft. Use the porosity relationship determined in Example 1.

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Compact Effect

$g = g_o + \Delta g$

$\Delta g = g_o - g_{DS}$

K

$\phi_1 = e^{-KDS}$

$0.052 = 2.6 - 1.074 \ln(8.33 - 0.408)$

$\phi = 0.052 - 2.6 \ln(8.33 - 10,000)$

0.0000856

$\phi =$

$1 - e^{-0.0000856 \times 10,000} = 9,436 \text{ psi}$

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Differential Density Effects

This effect is encountered when a gas reservoir with a significant dip is drilled. Because of a failure to recognize this potential hazard, blowouts may occur.

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Differential Density Effects

Example 3: Consider the gas sand shown in Figure 1.2, which was encountered in the U.S. gulf coast area. If the water-filled portion of the

sand is pressured normally and the gas/water contact occurred at a depth of 5000 ft, what mud weight would be required to drill through the top of the sand structure safely at a depth of 4000 ft? Assume the gas has an average density of 0.8 lbm/gal.

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Differential Density Effects

Solution: $P_{5000ft} = P_{4000ft} + P_{Gas1000ft}$

$P_{4000ft} = P_{5000ft} - P_{Gas1000ft}$

$P_{4000ft} = 0.465(\text{psi/ft}) \times 5000 (\text{ft}) - 0.052 \times 0.8 (\text{lbm/gal}) \times 1000 (\text{ft})$

$P_{4000ft} = 2283 \text{ psi.}$

The mud density needed to balance this pressure while drilling

p 4000 ft 2283

$\rho_m = 11 \text{ lbm / gal}$

$0.052 \times h = 0.052 \times 4000$

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Estimation of Abnormal Formation Pressure

The predictive techniques are based on measurements that can be made:

1. Geophysical measurements: identify geological conditions which might indicate the potential for overpressures such as salt domes
2. Analyzing data from wells that have been drilled in nearby locations (offset wells).
3. Seismic data has been used successfully to identify transition zones
4. Offset well histories may contain information on mud weights used, problems with stuck pipe, lost circulation or kicks.
5. Wireline logs or mudlogging information is also valuable when attempting to predict overpressures.

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Estimation of Abnormal Formation Pressure

Based on Drilling Parameters

The theory behind using drilling parameters to detect overpressured zones is based on the fact that:

1. Compaction of formations increases with depth. ROP will therefore, all other things being constant, decrease with depth
2. In the transition zone the rock will be more porous (less compacted) than that in a normally compacted formation and this will result in an increase in ROP. Also,

as drilling proceeds, the differential pressure between the mud hydrostatic and formation pore pressure in the transition zone will reduce, resulting in a much greater ROP.

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Estimation of Abnormal Formation Pressure

Based on Drilling Parameters

Torque can be useful for identifying overpressured zones. An increase in torque may occur of the decrease in overbalance results in the physical breakdown of the borehole wall and more material, than the drilled cuttings is accumulating in the annulus. There is also the suggestion that the walls of the borehole may squeeze into the open hole as a result of the reduction in differential pressure. Drag may also increase as a result of these effects, although increases in drag are more difficult to identify.

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Estimation of Abnormal Formation Pressure

Based on Drilling Parameters

The use of the ROP to detect transition and therefore overpressured zones is a simple concept, but difficult to apply in practice. This is due to the fact that many factors affect the ROP, apart from formation pressure (e.g. rotary speed and WOB). Since these other effects cannot be held constant, they must be considered so that a direct relationship between ROP and formation pressure can be established. This is achieved by applying empirical equations to produce a “normalised” ROP, which can then be used as a detection tool.

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Estimation of Abnormal Formation Pressure

Based on Drilling Parameters

The ROP usually changes significantly with formation type. Therefore, the ROP log is one of the important factors to predict formation pressure.

The ROP is a function of many factors other than the formation type and formation pressure including: bit size, bit diameter, bit nozzle sizes, WOB, RPM, mud type, mud density, rheology of mud, pump pressure, pump rate. Therefore, it is difficult to detect formation pressure changes using only ROP

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Based on Drilling Parameters

Estimation of Abnormal
Formation Pressure

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Estimation of Abnormal Formation Pressure

Based on Drilling Parameters

Based on the considerable laboratory and field data, Bingham suggested an equation to calculate the ROP

a_5

$R = K \frac{W^{a_5}}{db^2 N}$

R = ROP, K = constant, N = rotary speed

db = bit diameter

where W is the bit weight, db is the bit diameter, N is the rotary speed, a_5 is the bit weight exponent and K is the constant of proportionality that includes the effect of rock strength

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Estimation of Abnormal Formation Pressure

Jorden and Shirley Model

Jorden and Shirley proposed using the Bingham model to normalize penetration rate R through the calculation of a d -exponent defined by

$R = \frac{1}{d \exp}$

$\log \frac{R}{R_0} = \frac{d \exp}{60} \log \frac{W}{W_0}$

$d \exp = \frac{60}{\log \frac{R}{R_0}} \log \frac{W}{W_0}$

$\log \frac{R}{R_0} = \frac{d \exp}{60} \log \frac{W}{W_0}$

$\log \frac{R}{R_0} = \frac{d \exp}{60} \log \frac{W}{W_0}$

$\log \frac{R}{R_0} = \frac{d \exp}{60} \log \frac{W}{W_0}$

The $d \exp$ can be used to detect the transition from normal to abnormal pressure if the drilling fluid density is held constant.

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Estimation of Abnormal Formation Pressure

Rehm and McClendon Model

Rehm and McClendon proposed modifying the $d \exp$ to correct for the effect of mud density changes as well as changes in WOB, bit diameter, and rotary speed.

n

$d_{mod} \neq d_{exp}$

ρ_e

where ρ_n is the mud density equivalent to a normal pore pressure gradient and ρ_e is the equivalent mud density at the bit while circulating

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Estimation of Abnormal Formation Pressure

Modified d-exponent

data in U.S. Gulf

Coast shales

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Estimation of Abnormal Formation Pressure

Rehm and McClendon Model

Example 4: A penetration rate of 23 ft/hr was observed while drilling in shale at a depth of 9,515 ft using a 9.875-in bit in the U.S. gulf coast area. The WOB was 25,500 lbf and the rotary speed was 113 RPM. The equivalent circulating density at the bit was 9.5 lbm/gal. Compute the d_{exp} and the d_{mod} . The normal pressure gradient in the U.S. gulf coast is 0.465 psi/ft.

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Estimation of Abnormal Formation Pressure

Rehm and McClendon Model

$R = 23$

$\log R \neq \log R$

$60 N = 60 \times 113 = 1.64$

d_{exp}

$12W = 12 \times 25,500$

$\log \log \log$

$1,000 d_b = 1,000 \times 9.875$

0.465

$\rho_n = 8.94 \text{ lbm / gal}$

0.052

$\rho_n = 8.94$

$d_{mod} \neq d_{exp} = 1.64 \neq 1.54$

$\rho_e = 9.5$

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Estimation of Abnormal Formation Pressure

Rehm and McClendon Model

The modified dexp often is used for estimating the formation pressure gradient as well as the abnormal formation pressure. Rehm and McClendon suggested the following empirical equation to calculate the equivalent mud density

$$\rho_e = 7.65 \log \left(\frac{\rho_d}{\rho_n} \right) + \rho_{abn} + 16.5$$

$$\text{Formation pressure: } P_f = 0.052 \rho_e$$

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Estimation of Abnormal Formation Pressure

Zamora Model

Zamora also introduced another empirical equation to calculate the formation pressure gradient

$$\rho_d = \rho_n$$

$$(gf)_a = (gf)_n$$

$$\rho_d = \rho_{abn}$$

Where $(gf)_a$ and $(gf)_n$ – abnormal formation pressure gradient and normal formation pressure gradient, psi/ft

$$\text{The abnormal formation pressure: } P_f = (gf)_a \times D$$

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Estimation of Abnormal Formation Pressure

Example: Given dexp vs. depth as shown in the Figure. Estimate the formation pressure at 13,000 ft using Rehm and McClendon and the Zamora correlation.

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Estimation of Abnormal Formation Pressure

Rehm and McClendon Method

Equivalent density

$$\rho_e = 7.65 \log \left(\frac{\rho_d}{\rho_n} \right) + \rho_{abn} + 16.5$$

$$\rho_e = 7.65 \log \left(\frac{1.64}{1.17} \right) + 16.5 = 14 \text{ lbm / gal}$$

Formation pressure gradient

$$P_f = 0.052 \times e$$

$$P_f = 0.052 \times e = 0.052 \times 14 = 0.728 \text{ psi / ft}$$

Formation pressure at 13,000 ft

$$P_f(13,000\text{ft}) = 9,464 \text{ psi}$$

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Estimation of Abnormal Formation Pressure

Zamora method

$$g = g_{mod} \times n$$

$$f_a = f_n$$

$$d_{mod} = d_{abn}$$

$$g = 0.465 \times 1.64 = 0.652 \text{ psi / ft}$$

$$f_a$$

$$1.17$$

$$P_f(13,000) = 0.652 \times 13,000 = 8476 \text{ psi}$$

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Detection of Formation Pressure

Based on Seismic Data

To estimate formation pore pressure from seismic data, the average acoustic velocity as a function of depth must be determined. A geophysicist who specializes in computer assisted analysis of seismic data usually performs this for the drilling engineer. For convenience, the reciprocal of velocity or interval transit time, generally is displayed.

Interval transit time is the amount of time for a wave to travel a certain distance, proportional to the reciprocal of velocity, typically measured in microseconds per foot by an acoustic log and symbolized by t . The acoustic log displays travel time of acoustic waves versus depth in a well. The term is commonly used as a synonym for a sonic log. Some acoustic logs display velocity.

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Detection of Formation Pressure

Based on Seismic Data

The relationship between the interval transit time t and porosity:

$$t = t_{ma}(1 - \phi) + t_{fl}\phi$$

where t_{ma} is the interval transit time in the rock matrix and t_{fl} is the interval transit time in the pore fluid. Since transit times are greater for fluids than for solids, the

observed transit time in rock increases with increasing porosity.

With $\alpha = \alpha_{oe}$ -KDs

$$t = t_{ma}(1 - \alpha_{oe}\text{-KDs}) + t_{fl} \alpha_{oe}\text{-KDs}$$

$$t = t_{ma} + \alpha_o(t_{fl} - t_{ma})e\text{-KDs}$$

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Detection of Formation Pressure

Based on Seismic Data

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Detection of Formation Pressure

Based on Seismic Data

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Detection of Formation Pressure

Based on Seismic Data

Example:

The average interval transit time data shown in Table 6.4 were computed from seismic records of normally pressured sediments occurring in the upper miocene trend of the Louisiana gulf coast. These sediments are known to consist mainly of sands and shales. Using these data and the values of K and α_o computed previously for the U.S. gulf coast area in Example 6.2, compute apparent average matrix travel times for each depth interval given and curve fit the resulting values as a function of porosity. A water salinity of approximately 90,000 ppm is required to give a pressure gradient of 0.465 psi/ft.

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Detection of Formation Pressure

Based on Seismic Data

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Detection of Formation Pressure

Based on Seismic Data

Solution:

The values of α_o and K determined for the US gulf coast area in Example 6.2 were 0.41 and 0.000085 1/ft, respectively. From Table 6.3, a value of 209 is indicated for interval transit time in 90,000-ppm brine.

$$\alpha = 0.41e-0.000085D$$

$$t_{ma} = (t - 209\alpha) / (1 - \alpha)$$

From these two equations, for any given depths, we should be able to calculate the average porosity and interval transit time of the rock matrix

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Detection of Formation Pressure

Based on Seismic Data

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Detection of Formation Pressure

Based on Seismic Data

$$t_{ma} = 50 + 180\alpha. \text{ Substitute this equation to: } t = t_{ma}(1 - \alpha) + t_{fl}\alpha \text{ with } t_{fl} = 209$$

$$t = 209\alpha + (50 + 180\alpha)(1 - \alpha)$$

$$t = 50 + 339\alpha - 180\alpha^2$$

$$\text{With } \alpha = 0.41e-0.000085D$$

$$t = 50 + 339\alpha e^{-0.000085D} - 180(\alpha e^{-0.000085D})^2$$

Average interval transit time depends only on the surface porosity, porosity constant decline K and the depth, D.

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Detection of Formation Pressure

Based on Seismic Data

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Detection of Formation Pressure

Based on Seismic Data

Example: The average interval transit time data shown in Table 6.6 were computed from seismic records at a proposed well location in the south Texas Frio trend. Estimate formation pressure at 9,000 ft. Extend the mathematical model for the normal pressure trend developed in the previous example to this trend; select an appropriate value of average surface porosity, α_0 .

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Detection of Formation Pressure

Based on Seismic Data

The first method that can be used to estimate formation pressure at 9,000 ft is an

empirically determined relationship between interval transit time and formation pressure. The ratio of observed transit time to normal interval transit time at 9000 ft is

$$t / t_n = 129 / 92 = 1.4$$

From the graph, the formation pore pressure gradient is 0.93 psi/ft. The formation pressure is

$$P = 0.93 \times 9,000 = 8,370 \text{ psig.}$$

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Detection of Formation Pressure

Based on Seismic Data

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Detection of Formation Pressure

Based on Seismic Data

Rearrange this equation: $t_n = 50 + 339 \phi_o e^{-0.000085D} - 180 \phi_o^2 e^{-0.00017D}$ to calculate the surface porosity gives

With $D = 2000$ ft and the interval transit time 137, $\phi_o = 0.364$. Repeat the calculation with different depths, the results are shown in Table:

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Detection of Formation Pressure

Based on Seismic Data

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Detection of Formation Pressure

Based on Seismic Data

Louisiana

gulf coast

South

Texas

Frio

Trend

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Detection of Formation Pressure

Based on Seismic Data

The average surface porosity is 0.285. Thus the normal pressure trend line equation becomes:

$$t_n = 50 + 96.6e^{-0.000085D} - 14.6e^{-0.00017D}$$

The second approach that can be used to estimate formation pressure at 9000 ft is based on the assumption that formations having the same value of interval transit time are under the same vertical effective matrix stress, σ_z . At 9,000 ft, the interval transit time has a value of 129. The depth of the normally pressured formation having this same value of interval transit time

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Detection of Formation Pressure

Based on Seismic Data

The vertical overburden stress, σ_{ob} at the depth of 1300:

$$0.052 \text{ g/cc} \times 1300 \text{ ft}$$

$$(\sigma_{ob})_{1,300} = 0.052 \text{ g/cc} \times 1300 \text{ ft}$$

K

$$1 \text{ e}^{-0.000085 \times 1300} = 0.88$$

$$0.052 \times 2.6 \times 1.074 \times 8.33 \times 0.285$$

$$(\sigma_{ob})_{1,300} = 0.052 \text{ g/cc} \times 2.6 \times 8.33 \times 1,300 \text{ ft}$$

$$0.000085$$

σ_z

$$1 \text{ e}^{-0.000085 \times 1,300} = 1,232 \text{ psig}$$

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Detection of Formation Pressure

Based on Seismic Data

The vertical overburden stress, σ_{ob} at the depth of 1300:

The formation pressure at 1,300 ft is given by:

$P_{1,300\text{ft}} = 0.465 \times 1,300 = 605 \text{ psig}$. Thus the effective stress at both 1,300 and 9,000 ft is

$$\sigma_{9,000} = \sigma_{1,300} = (\sigma_{ob})_{1,300} - P_{1,300} = 1,232 - 605 = 627 \text{ psig}.$$

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Detection of Formation Pressure

Based on Seismic Data

The vertical overburden stress, σ_{ob} at the depth of 9,000:

$$0.052 \text{ g/cc} \times 9,000 \text{ ft}$$

$$(\sigma_{ob})_{9,000} = 0.052 \text{ g/cc} \times 9,000 \text{ ft}$$

K

$\frac{1}{1000} \times \text{KDS}$

$0.052 \times 2.6 \times 1.074 \times 8.33 \times 0.285$

$(\frac{1}{1000} \times V) \times 9,000 \times 0.052 \times 2.6 \times 8.33 \times 9,000 \times$

0.000085

$\frac{1}{1000}$

$1 \times \frac{1}{1000} \times 0.000085 \times 9,000 \times 8,951 \text{ psig}$

Thus, the pore pressure at 9,000 ft:

$P_{9,000} = (P_{\text{ob}})_{9,000} - \frac{1}{1000} \times 9,000 = 8,951 - 627 = 8,324 \text{ psig}$.

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Detection of Formation Pressure

Based on Drilling Mud Parameters

The main effects on the mud due to abnormal pressures will be:

1. Increasing gas cutting of mud
2. Decrease in mud weight
3. Increase in flowline temperature

Since these effects can only be measured when the mud is returned to surface

they involve a time lag of several hours in the detection of the overpressured zone.

During the time it takes to circulate bottoms up, the bit could have penetrated quite far into an overpressured zone.

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Detection of Formation Pressure

Based on Drilling Mud Parameters

(a) Gas Cutting of Mud

Gas cutting of mud may happen in two ways:

1. From shale cuttings: if gas is present in the shale being drilled the gas may be released into the annulus from the cuttings.
2. Direct influx: this can happen if the overbalance is reduced too much, or due to swabbing when pulling back the drillstring at connections.

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Detection of Formation Pressure

Based on Drilling Mud Parameters

(b) Mud Weight

The mud weight measured at the flowline will be influenced by an influx of

formation fluids. The presence of gas is readily identified due to the large decrease in density, but a water influx is more difficult to identify. Continuous measurement of mud weight may be done by using a radioactive densometer.

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(c) Flowline Temperature

Under-compacted clays, with relatively high fluid content, have a higher temperature than other formations. By monitoring the flowline temperature therefore an slow increase in temperature will be observed when drilling through normally pressured zones. This will be followed by an rapid increase in temperature when the overpressured zones are encountered. The normal geothermal gradient is about 1 degree F/100 ft. It is reported that changes in flowline temperature up to 10 degree F/100 ft. have been detected when drilling overpressured zones.

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Detection of Formation Pressure

Based on Drilled Cuttings

Since overpressured zones are associated with under-compacted shales with high fluid content the degree of overpressure can be inferred from the degree of compaction of the cuttings. The methods commonly used are:

1. Density of shale cuttings
2. Shale factor
3. Shale slurry resistivity

Even the shape and size of cuttings may give an indication of overpressures (large cuttings due to low pressure differential). As with the drilling mud parameters these tests can only be done after a lag time of some hours.

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Detection of Formation Pressure

Based on Drilled Cuttings

(a) Density of Shale Cuttings

In normally pressured formations the compaction and therefore the bulk density of shales should increase uniformly with depth (given constant lithology). If the bulk density decreases, this may indicate an undercompacted zone which may be an overpressured zone. The bulk density of shale cuttings can be determined by using a mud balance.

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(b) Shale Factor

This technique measures the reactive clay content in the cuttings. It uses the “methylene blue” dye test to determine the reactive montmorillonite clay present, and thus indicate the degree of compaction. The higher the montmorillonite, the lighter the density - indicating an undercompacted shale.

Montmorillonite will absorb methylene blue and change its color.

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(c) Shale Slurry Resistivity

As compaction increases with depth, water is expelled and so conductivity is reduced. A plot of resistivity against depth should show a uniform increase in resistivity, unless an undercompacted zone occurs where the resistivity will reduce. To measure the resistivity of shale cuttings a known quantity of dried shale is mixed with a known volume of distilled water. The resistivity can then be measured and plotted

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